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A novel SUN1-ALLAN complex coordinates segregation of the bipartite MTOC across the nuclear envelope during rapid closed mitosis in Plasmodium

Mohammad Zeeshan, Igor Blatov, Ryuji Yanase, David JP Ferguson, Sarah L Pashley, Zeinab Chahine, Yoshiki Yamaryo Botté, Akancha Mishra, Baptiste Marché, Suhani Bhanvadia, Molly Hair, Sagar Batra, Robert Markus, Declan Brady, Andrew Bottrill, Sue Vaughan, Cyrille Y Botté, Karine Le Roch, Anthony A Holder, Eelco C Tromer, Rita Tewari

University of Nottingham, School of Life Sciences, Nottingham, United Kingdom • Oxford Brookes University, Department of Biological and Medical Sciences, Oxford, United Kingdom • Department of Molecular, Cell and Systems Biology, University of California Riverside, Riverside, United States • Apicolipid Team, Institute for Advanced Biosciences, CNRS UMR5309, Université Grenoble Alpes, INSERM U1209, Grenoble, France • School of Life Sciences, Gibbet Hill Campus, University of Warwick, Coventry, United Kingdom • Malaria Parasitology Laboratory, The Francis Crick Institute, London, United Kingdom • Cell Biochemistry, Groningen Biomolecular Sciences and Biotechnology Institute, Faculty of Science and Engineering, University of Groningen, Groningen, Netherlands

eLife Assessment

In Plasmodium male gametocytes, rapid nuclear division occurs with an intact nuclear envelope, requiring precise coordination between nuclear and cytoplasmic events to ensure proper packaging of each nucleus into a developing gamete. This **valuable** study characterizes two proteins involved in the formation of Plasmodium berghei male gametes. By integrating live-cell imaging, ultrastructural expansion microscopy, and proteomics, this study **convincingly** identifies SUN1 and its interaction partner ALLAN as crucial nuclear envelope components in male gametogenesis. A role for SUN1 in membrane dynamics and lipid metabolism is less well supported. The results are of interest for general cell biologists working on unusual mitosis pathways.

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Abstract

Mitosis in eukaryotes involves reorganization of the nuclear envelope (NE) and microtubule-organizing centres (MTOCs). During male gametogenesis in Plasmodium, the causative agent of malaria, mitosis is exceptionally rapid and highly divergent. Within 8 min, the haploid male gametocyte genome undergoes three replication cycles (1N to 8N), while maintaining an intact NE. Axonemes assemble in the cytoplasm and connect to a bipartite MTOC-containing nuclear pole (NP) and cytoplasmic basal body, producing eight flagellated gametes. The mechanisms coordinating NE remodelling, MTOC dynamics, and flagellum assembly remain poorly understood.

We identify the SUN1-ALLAN complex as a novel mediator of NE remodelling and bipartite MTOC coordination during *Plasmodium* male gametogenesis. SUN1, a conserved NE protein, localizes to dynamic loops and focal points at the nucleoplasmic face of the spindle poles. ALLAN, a divergent allantoicase, has a location like that of SUN1, and these proteins form a unique complex, detected by live-cell imaging, ultrastructural expansion microscopy, and interactomics. Deletion of either SUN1 or ALLAN genes disrupts nuclear MTOC organization, leading to basal body mis-segregation, defective spindle assembly, and impaired spindle microtubule-kinetochore attachment, but axoneme formation remains intact. Ultrastructural analysis revealed nuclear and cytoplasmic MTOC miscoordination, producing aberrant flagellated gametes lacking nuclear material. These defects block development in the mosquito and parasite transmission, highlighting the essential functions of this complex.

Introduction

Mitosis, the process of eukaryotic cell division, requires the accurate segregation of nuclear and cytoplasmic materials, coordinated by dynamic changes in the nuclear envelope (NE) and microtubule-organizing centres (MTOCs) (Liu and Pellman, 2020). The NE, a double-membrane structure with embedded nuclear pore (NP) complexes, is a selective barrier, facilitating signal exchange between cytoplasm and nucleus. The NE consists of inner (INM) and outer (ONM) membranes, with the ONM continuous with the endoplasmic reticulum (ER). Beyond structural and transport roles, the NE is integral to mitosis, supporting spindle formation, kinetochore attachment, and chromosome segregation, while also accommodating changes in ploidy (Dey and Baum, 2021; Smoyer and Jaspersen, 2014).

In most mammalian cells, mitosis is considered ‘open’, characterized by complete NE disassembly, chromosome condensation and segregation via a microtubule-based bipolar spindle. In contrast, in unicellular organisms such as the budding yeast *Saccharomyces cerevisiae*, mitosis is comparatively ‘closed’: the NE remains intact, and chromosomes are segregated by an intranuclear spindle anchored to acentriolar NPs (Boettcher and Barral, 2013; Sazer et al., 2014). However, it is now believed that no mitosis is wholly ‘open’ or ‘closed’ and that in most cases remnants of the NE interact with the spindle to support chromosome segregation (Dey and Baum, 2021). Closed mitosis involves diverse NE dynamics: for example, *Trypanosoma* and *Saccharomyces* employ intranuclear spindle assembly, whereas *Chlamydomonas* and *Giardia* assemble spindles outside the nucleus; the NE remains largely unbroken, but large polar fenestrae are formed to allow microtubule access to the chromosomes (Makarova and Oliferenko, 2016). These examples highlight the fact that ‘open’ and ‘closed’ mitosis are merely two extremes of a continuum of NE remodelling strategies during mitosis, underlining extensive diversity in cell division, a core cellular process.

The interaction between nucleus, the chromatin and in general the nucleolplasmic environment and the cytoskeleton is facilitated by the LINC (Linker of Nucleoskeleton and Cytoskeleton) complex (**Fig. 1**), which bridges the INM and ONM in most eukaryotes via interactions between SUN (Sad1, UNC84)-domain proteins on the INM and KASH (Klarsicht, ANC-1, Syne Homology)-domain proteins on the ONM (Hao and Starr, 2019; Starr and Fridolfsson, 2010). The LINC complex links chromatin and the cytoskeleton throughout eukaryotes (**Fig. 1**). For instance, LINC complexes anchor telomeres to the NE during meiosis in budding yeast (Schober et al., 2008) and cluster centromeres near the INM in fission yeast (Funabiki et al., 1993). Furthermore, SUN proteins connect to the lamin-like proteins that cover the INM on the nucleoplasmic side in animals and plants (Koreny and Field, 2016). In plants, SUN–WIP complexes (analogous to the SUN-KASH proteins of animals and fungi) connect the nucleus to actin filaments (Zhou et al., 2015), while Dictyostelium uses SUN1 to anchor the spindle pole body-like MTOC at the NE (Xiong et al., 2008). Such interactions coordinate nuclear positioning, chromosome organization, and mechanical integration with the cytoskeleton. Evolutionary studies suggest that LINC was established before the emergence of the last eukaryotic common ancestor (LECA), highlighting its ancient role in NE formation and cytoskeletal coordination (**Fig. 1**) (Baum and Baum, 2014; Koreny and Field, 2016). However, as SUN proteins are broadly conserved in many eukaryotic lineages, KASH and lamin-like proteins are often not detected (Koreny and Field, 2016). Intriguingly, in many non-model eukaryotes like for instance apicomplexan parasites (e.g. plasmodium and toxoplasma) microscopic observations reveal chromatin structures such as centromeres, telomeres and the spindle pole body-like MTOCs are to be closely associated with and/or embedded in the NE (**Fig. 1**). *Have KASH/lamins proteins been lost or have novel systems evolved to support a SUN protein-based functions at the nuclear envelope?*

The malaria-causing parasite *Plasmodium spp.* uses closed mitosis with some highly divergent features across its complex life cycle. In the asexual stage in the blood of its vertebrate host, during schizogony there is asynchronous nuclear division without coincident cytokinesis, and during male gametogenesis in the mosquito host there is a unique and rapid form of closed mitosis essential for parasite transmission (Guttery et al., 2022; Sinden et al., 1978). Following activation in the mosquito gut, the haploid male gametocyte undergoes three rounds of genome replication (from 1N to 8N) within just 6 to 8 min, while maintaining an intact NE. Concurrently, axonemes form in the cytoplasm, emanating from bipartite MTOCs that consist of intranuclear spindle poles and cytoplasmic basal bodies (BB). This rapid karyokinesis and subsequent cytokinesis results in the production of eight flagellated haploid male gametes within 15 min (Guttery *et al*., 2022).

The speed of male gametogenesis in *Plasmodium* imposes unique requirements and constraints on cellular structures. Notably, the flagella lack intraflagellar transport (IFT), which is atypical (Sinden et al., 2010). The bipartite organization of the MTOC was recently revealed, and use of fluorescently tagged markers such as SAS4 and kinesin-8B has illuminated the dynamics of BB and axoneme formation (Zeeshan et al., 2022a; Zeeshan et al., 2019a). Kinetochore proteins like NDC80 display an unconventional, largely clustered linear organization on the spindle, redistributing only during successive spindle duplication (Zeeshan et al., 2020). Remarkably, successive spindle and BB segregation occurs within 8 min of gametocyte activation, without NE breakdown, indicating an unusual, streamlined closed mitotic process (Zeeshan *et al*., 2022a; Zeeshan *et al*., 2019a).

The mechanisms that coordinate the formation and function of the intranuclear spindle poles, the cytoplasmic BB, and the NE remain unclear. Specifically, how the NE is remodelled to accommodate rapid genome replication, and how it affects the organization and function of the bipartite MTOC remain open questions. We addressed these questions by investigating NE remodelling and MTOC coordination in *Plasmodium berghei* (Pb), using the conserved NE protein SUN1. By combining live-cell imaging, ultrastructural expansion microscopy (U-ExM), and proteomic analysis, we identify SUN1 as a key NE component. Using different mitotic and

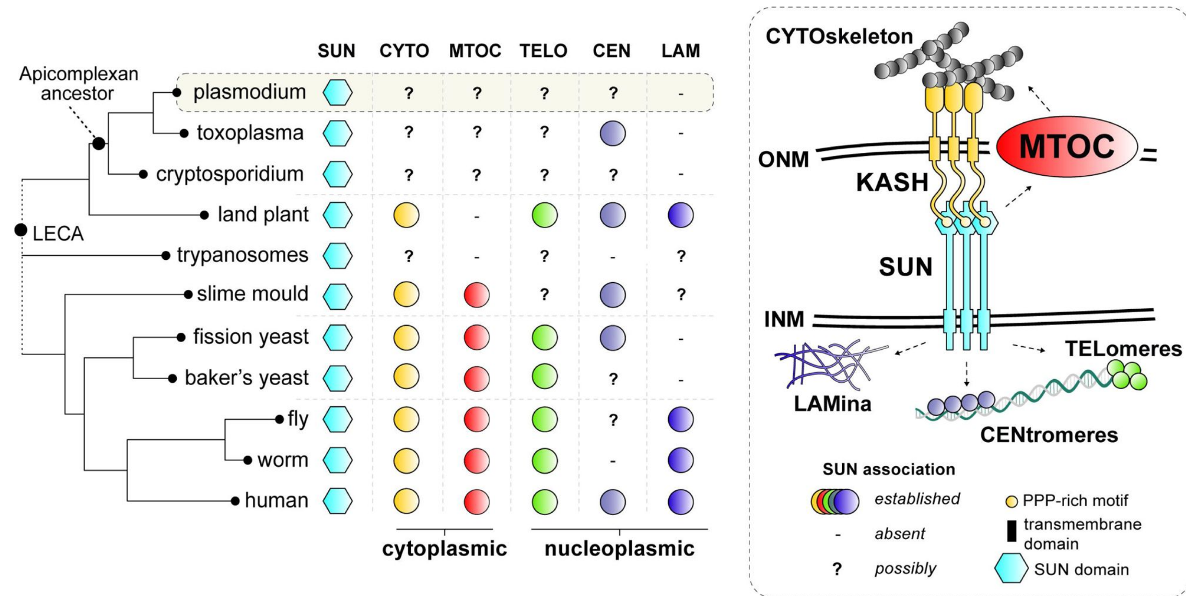


Fig 1.

Comparative analysis of SUN protein functions in common eukaryotic model systems.

SUN proteins bridge the outer (ONM) and inner (INM) membranes of the nuclear envelope (NE) to link the cytoskeleton (i.e. actin, microtubules and its organising centres) to various heterochromatic domains (i.e. centromere, telomeres) and the nuclear lamina (five different functions with a unique colour). Established roles of SUN proteins in common model organisms are depicted by coloured circles. '-' no functional connection for SUN was found and/or structures are not present. '?' denotes possible roles for SUN proteins as NE connections with such structures have been established in these lineages. LECA: Last Eukayotic Common Ancestor.

MTOC/BB markers, we investigate their coordination and the flexibility of NE during rapid mitosis. Our findings reveal that SUN1 interacts with a novel allantoicase-like protein (termed ALLAN), to form a divergent LINC-like complex without KASH proteins. Functional disruption of either SUN1 or ALLAN impairs BB segregation, disrupts spindle-kinetochore attachment, and results in defective flagellum assembly. Our results highlight a unique divergence of NE remodelling and MTOC organization in *Plasmodium* gametogenesis from conventional eukaryotic models system found in animals, fungi and plants.

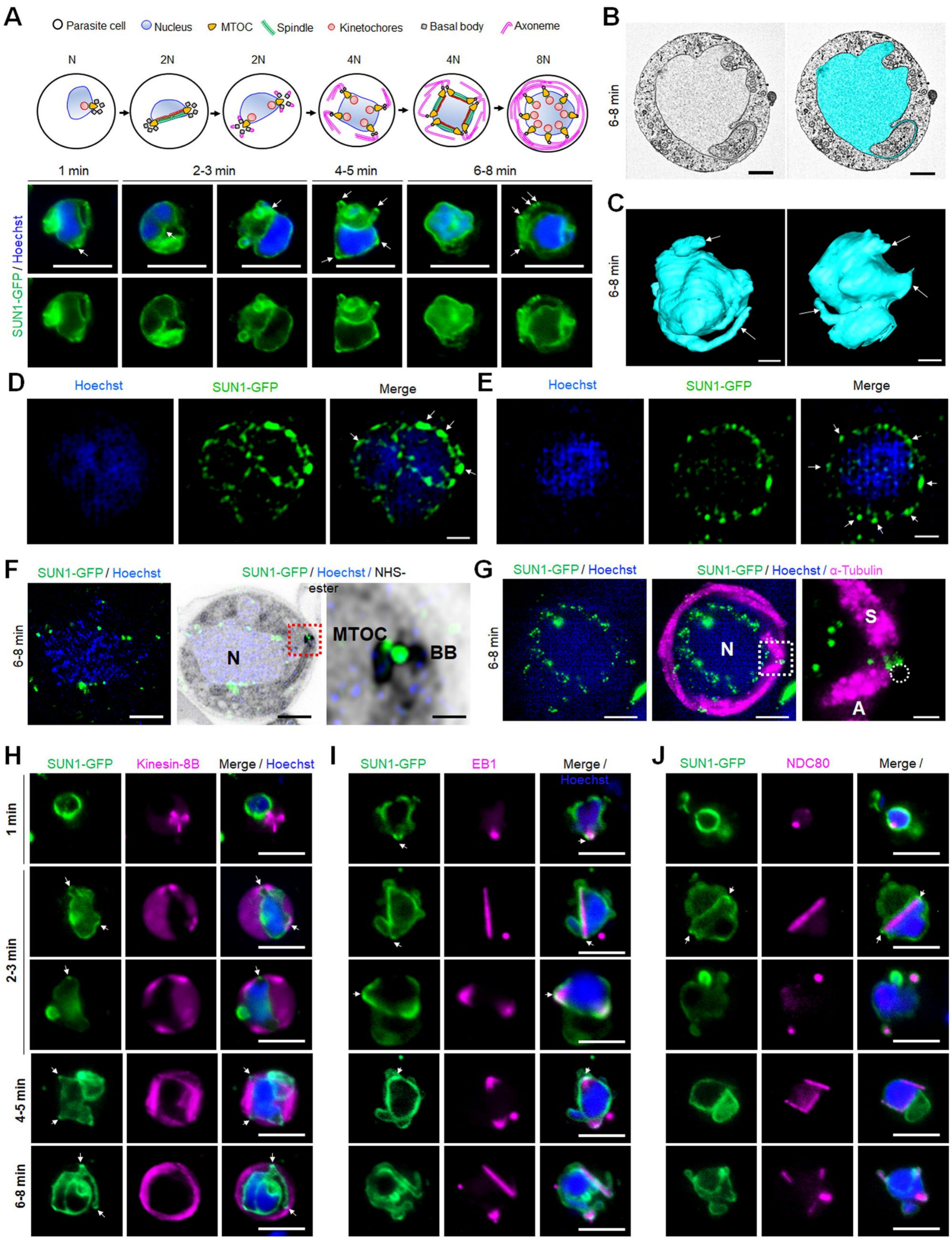
Results

Generation and Validation of PbSUN1 Transgenic Lines

To investigate the role of PbSUN1 (PBANKA_1430900) during *P. berghei* male gametogenesis, we generated transgenic parasite lines expressing a C-terminal GFP-tagged SUN1 protein (SUN1-GFP). The GFP-tagging construct was integrated into the 3' end of the endogenous *sun1* locus via single-crossover recombination (Fig. S1A), and correct integration was confirmed by PCR analysis using diagnostic primers (Fig. S1B). Western blot analysis detected SUN1-GFP at the expected size (~130 kDa) in gametocyte lysates (Fig. S1C). The SUN1-GFP line grew normally and progressed through the life cycle, indicating that tagging did not disrupt PbSUN1 function. The SUN1-GFP line was used to study the subcellular localization of PbSUN1 across multiple life cycle stages, and its interaction with other proteins during gametogenesis.

Spatiotemporal Dynamics of SUN1 During Male Gametogenesis

SUN1 protein expression was undetectable by live-cell imaging during asexual erythrocytic stages of the parasite, but robust expression was visible in both male and female gametocytes following activation in ookinete medium (Fig. S1D, E). In activated male gametocytes undergoing mitotic division, SUN1-GFP had a dynamic localization around the nuclear DNA (Hoechst-stained), associated with loops and folds formed in the NE during the transition from 1N to 8N ploidy (**Fig. 2A**, Supplementary Video 1). The NE loops extended beyond the Hoechst-stained DNA, suggesting an abundant non-spherical NE membrane (**Fig. 2A**). To further characterize NE morphology during male gametogenesis, serial block-face scanning electron microscopy (SBF-SEM) was used. Analysis of wild-type male gametocytes revealed a non-spherical, contorted nucleus (**Fig. 2B**, S1F). Thin loops of NE (indicated with arrows) were prominent, consistent with the dynamic and irregular location of SUN1-GFP detected by live-cell imaging. 3D-modelling of gametocyte nuclei further confirmed the irregular, non-spherical structure of the NE (**Fig. 2C**, S1G), which expanded rapidly during the period of genome replication. These findings suggest that the male gametocyte NE is a highly dynamic structure that can expand rapidly to accommodate the increased DNA due to replication during gametogenesis.


A
Parasite cell
Nucleus
MTOC
Spindle
Kinetochores
Basal body
Axoneme
N
2N
2N
4N
4N
8N
1 min
2-3 min
4-5 min
6-8 min
SUN1-GFP / Hoechst
B
6-8 min
C
6-8 min
D
Hoechst
SUN1-GFP
Merge
E
Hoechst
SUN1-GFP
Merge
F
SUN1-GFP / Hoechst
SUN1-GFP / Hoechst / NHS-ester
6-8 min
N
MTOC
BB
G
SUN1-GFP / Hoechst
SUN1-GFP / Hoechst / α-Tubulin
6-8 min
N
S
A
H
SUN1-GFP
Kinesin-8B
Merge / Hoechst
1 min
2-3 min
4-5 min
6-8 min
I
SUN1-GFP
EB1
Merge / Hoechst
J
SUN1-GFP
NDC80
Merge /

[Fig 2.](#)

Location of SUN1 during male gametogenesis.

A. The upper panel schematic illustrates the process of male gametogenesis. N, genome ploidy. Live cell images show the location of SUN1-GFP (green) at different time points(1-8min) during male gametogenesis. DNA(blue) was stained with Hoechst. White arrows indicate the loop/folds. Representative images of more than 50 cells with more than three biological replicates. Scale bar: 5 µm. **B.** Serial block face-scanning electron microscopy (SBF-SEM) data slice of a gametocyte highlighting the complex morphology of the nucleus (cyan). Representative of more than 10 cells. Scale bar: 1 µm. **C.** Two 3D models of gametocyte nuclei showing their contorted and irregular morphology. Representative of more than 10 cells. Scale bar: 1 µm. **D.** SIM images of SUN1-GFP male gametocytes activated for 8 min and fixed with paraformaldehyde. Arrows indicate the SUN1-GFP signals with high intensity after fixation. Representative image of more than 10 cells from more than two biological replicates. Scale: 1 µm. **E.** SIM images of SUN1-GFP male gametocytes activated for 8 min and fixed with methanol. Arrows indicate the SUN1-GFP signals with high intensity after fixation. Representative image of more than 10 cells from more than two biological replicates. Scale: 1 µm. **F.** Expansion microscopy (ExM) images showing location of SUN1 (green) detected with anti-GFP antibody and BB/MTOC stained with NHS ester (grey). Hoechst was used to stain DNA. Scale bar: 5 µm. Inset is the area marked with the red box around the BB/MTOC highlighted by NHS-ester staining. Scale bar: 1Ǉµm. Representative images of more than 10 cells from two biological replicates. **G.** ExM images showing location of SUN1 (green) and α-tubulin (magenta) detected with anti-GFP and anti-tubulin antibodies, respectively. Hoechst was used to stain DNA(blue). N=Nucleus; S=Spindle; A=Axoneme. Scale bar: 5 µm. Inset is the area marked with the white box on **Fig. 1E** middle panel around the BB/MTOC. Scale bar: 1Ǉµm. Representative images of more than 10 cells from two biological replicates. **H.** Live cell imaging showing location of SUN1-GFP (green) in relation to the BB and axoneme marker, kinesin-8B-mCherry (magenta) at different time points(1-5min) during gametogenesis. Blue in merged image is DNA stained with Hoechst. Representative images of more than 20 cells from more than three biological replicates. White arrows indicate the loops/folds labelled with SUN1 where BB/axonemes are assembled outside the nuclear membrane. Scale bar: 5Ǉµm. **I**. Live cell imaging showing location of SUN1-GFP (green) in relation to the spindle marker, EB1-mCherry (magenta) at different time points during gametogenesis. Blue in merged image is DNA stained with Hoechst. White arrows indicate the loops/folds labelled with SUN1. Representative images of more than 20 cells from more than three biological replicates. Scale bar: 5Ǉµm. **J**. Live cell imaging showing location of SUN1-GFP (green) in relation to the kinetochore marker, NDC80-mCherry (magenta) at different time points during gametogenesis. Blue in merged image is DNA stained with Hoechst. White arrows indicate the loops/folds labelled with SUN1. Representative images of more than 20 cells with more than three biological replicates. Scale bar: 5Ǉµm.

SUN1 Dynamics by High-Resolution Imaging

To resolve the SUN1-GFP location with higher precision, structured illumination microscopy (SIM) was performed on SUN1-GFP gametocytes fixed at 6-8 min post activation. Fixation of gametocytes caused SUN1-GFP signal coalescence into regions within the NE (Fig. S2A), but an uneven intensity of SUN1-GFP signal around the DNA was observed, with areas of higher intensity (**Fig. 2D, E**; S2B, white arrows) likely representing collapsed forms of the SUN1-GFP-labelled loops observed by live imaging (Fig. S2A). To refine the spatial relationship of SUN1 to the BB and the spindle, ultrastructure expansion microscopy (U-ExM) was used. At 6 to 8 min post-activation SUN1-GFP was prominently located around the DNA and at the junction of the nuclear MTOC and the BB (**Fig. 2F**; S3A). Spindle and axoneme staining with α-tubulin antibodies revealed that SUN1-GFP foci were located next to spindle poles (**Fig. 2G**; S3B), suggesting a role in MTOC organization during mitosis. WT-ANKA gametocytes (with no GFP present) did not react with the anti-GFP antibodies (Fig. S3C), confirming the specificity of the GFP signal in SUN1-GFP gametocytes.

The Location of SUN1 Relative to Key Mitotic Markers During Male Gametogenesis

To investigate the association of SUN1 with MTOCs, mitotic spindles, and BB/axonemes during male gametogenesis, its location was compared in real time with that of three markers: the cytoplasmic BB/axonemal protein kinesin-8B, the spindle microtubule-binding protein EB1, and the kinetochore marker NDC80 (Zeeshan *et al*., 2019a; Zeeshan *et al*., 2020; Zeeshan et al., 2023). A parasite line expressing SUN1-GFP (green) was crossed with lines expressing mCherry (red)-tagged kinesin-8B, EB1, or NDC80, and progeny were analyzed by live-cell imaging to determine the proteins' spatiotemporal relationships.

Within one-min post-activation, kinesin-8B appeared as a tetrad marking the BB in the cytoplasm and close to the nucleus, while SUN1-GFP localized to the nuclear membrane, with no overlap between the signals (**Fig. 2H**). As gametogenesis progressed, nuclear elongation was observed, with SUN1-GFP delineating the nuclear boundary. Concurrently, the BB (the kinesin-8B-marked tetrad) duplicated and separated to opposite poles in the cytoplasm, outside the SUN1-GFP-defined nuclear boundary (**Fig. 2H**). By 2 to 3 min post-activation, axoneme formation commenced at the BB, while SUN1-GFP marked the NE with small puncta located near the nuclear MTOCs situated between the BB tetrads (**Fig. 2H**).

Analysis of gametogenesis with SUN1-GFP and the spindle microtubule marker EB1, revealed a localized EB1-mCherry signal near the DNA, inside the nuclear membrane (marked by SUN1-GFP), within the first min post-activation (**Fig. 2I**). Loops and folds in the nuclear membrane, marked by SUN1-GFP, partially overlapped with the EB1 focal point associated with the spindle pole/MTOC (**Fig. 2I**, average Pearson Corelation Coefficient, R<0.6). At 2 to 3 min post activation, EB1 fluorescence extended to form a bridge-like spindle structure within the nucleus, flanked by two strong focal points that partially overlapped with SUN1-GFP fluorescence (**Fig. 2I**). A similar dynamic location was observed for another spindle protein, ARK2 (Fig. S4A).

Kinetochore marker NDC80 showed a similar pattern to that of EB1. Within one-min post-activation, NDC80 was detected as a focal point inside the nuclear membrane, later extending to form a bridge-like structure that split into two halves within 2 to 3 min, and with no overlap with SUN1-GFP (**Fig. 2J**). As with EB1, the nuclear membrane loops formed around NDC80 focal points, maintaining close proximity to the kinetochore bridge (**Fig. 2J**).

Together, these observations suggest that although SUN1-GFP fluorescence is partially colocated with spindle poles revealed by EB1 fluorescence, there is no overlap with kinetochores (NDC80) or BB/axonemes (kinesin-8B).

SUN1 Location During Female Gametogenesis and Zygote to Ookinete transformation

The location of SUN1-GFP during female gametogenesis was examined using real-time live-cell imaging. Within one-min post-activation, SUN1-GFP was observed to form a half-circle around the nucleus, eventually encompassing the entire nucleus after six to eight min (Fig. S4B). In contrast to SUN1-GFP in male gametocytes, SUN1-GFP in female gametocytes was more uniformly distributed around the nucleus without apparent loops or folds in the NE (Fig. S4B). This pattern probably reflects the absence of DNA replication and mitosis in female gametogenesis, so that the nucleus remains compact.

During zygote to ookinete differentiation and oocyst development, in zygotes SUN1-GFP had a spherical distribution around Hoechst-stained nuclear DNA at two hours post-fertilization (Fig. S4C), but by 24 hours post-fertilization, as the zygote developed into a banana-shaped ookinete, the

SUN1-GFP-marked NE became elongated or oval, possibly due to spatial constraints within the cytoplasm (Fig. S4C).

SUN1 is Essential for Basal Body Segregation and Axoneme-Nucleus Coordination During Male Gametogenesis

The role of SUN1 was assessed by deleting its gene using a double crossover homologous recombination strategy in a parasite line constitutively expressing GFP (WT-GFP) (Janse et al., 2006) (Fig. S5A). WT-GFP parasites express GFP at all stages of the life cycle, facilitating phenotypic comparisons. Diagnostic PCR confirmed the correct integration of the targeting construct at the *sun1* locus (Fig. S5B), and this was verified by qRT-PCR analysis, which showed complete deletion of the *sun1* gene in the resulting transgenic parasite (Δ*sun1*) (Fig. S5C).

Phenotypic analysis of the Δ*sun1* line was conducted across the life cycle, in comparison to a WT-GFP control. Two independent knockout clones (Clone-1 and Clone-4) were examined: the clones exhibited similar phenotypes, and one, Clone-4 was used for further experiments. Δ*sun1* parasites produced a comparable number of gametocytes to WT-GFP parasites, but had reduced male gametogenesis, as evidenced by a significant decrease in gamete formation (exflagellation) (**Fig. 3A**). The differentiation of zygotes into ookinetes was also reduced (**Fig. 3B**).

To evaluate sporogonic development, mosquitoes were fed with Δ*sun1* parasites, and oocyst formation was examined. The number of oocysts was markedly reduced in Δ*sun1* parasite-infected mosquitoes compared to WT-GFP parasites at days 7, 14, and 21 post-feeding (**Fig. 3C**). At day 7, oocysts were comparable in size but failed to grow further, and they had degenerated by day 21 (**Fig. 3D, E**) with no evidence of sporozoite production in midgut or salivary glands (**Fig. 3F, G**). Transmission experiments revealed that mosquitoes infected with Δ*sun1* parasites were unable to transmit the parasite to naïve mice; in contrast mosquitoes successfully transmitted the WT-GFP parasite, resulting in a blood-stage infection detected four days later (**Fig 3H**).

Because the defect in Δ*sun1* parasites led to a transmission block, we investigated whether the defect was rescued by restoring *sun1* into Δ*sun1* parasites using either the Δ*nek4* parasite that produces normal male gametocytes but is deficient in production of female gametocytes (Reininger et al., 2005) or the Δ*hap2* parasite that produces normal female gametocytes but is deficient in production of male gametocytes (Liu et al., 2008). We performed a genetic cross between Δ*sun1* parasites and the other mutants deficient in production of either male *(Δhap2)* or female *(Δnek4)* gametocytes. Crosses between Δ*sun1 and* Δ*nek4* mutants produced some normal-sized oocysts that were able to sporulate, showing a partial rescue of the Δ*sun1* phenotype (**Fig. 3I**). In contrast, crosses between Δ*sun1* and Δ*hap2* did not rescue the Δ*sun1* phenotype. As controls, Δ*sun1*, Δ*hap2* and Δ*nek4* parasites alone were examined and no oocysts/sporozoites were detected (**Fig. 3I**). To further confirm the viability of controls, male *(Δhap2)* and female *(Δnek4)* mutants were crossed together and produced normal-sized oocysts that were able to sporulate (**Fig 3I**). These results indicate that a functional *sun1* gene is required from a male gamete for subsequent oocyst development. To assess the effect of the *sun1* deletion on DNA replication during male gametogenesis, we analysed the DNA content (N) of Δ*sun1* and WT-GFP male gametocytes by fluorometric analyses after DAPI staining. We observed that Δ*sun1* male gametocytes were octaploid (8N) at 8 min post activation, similar to WT-GFP parasites (**Fig 3I**), indicating that the absence of SUN1 had no effect on DNA replication. We also checked for the presence of DNA in gametes stained with Hoechst (a DNA dye) and found that most Δ*sun1* gametes were anucleate (**Fig 3J**).

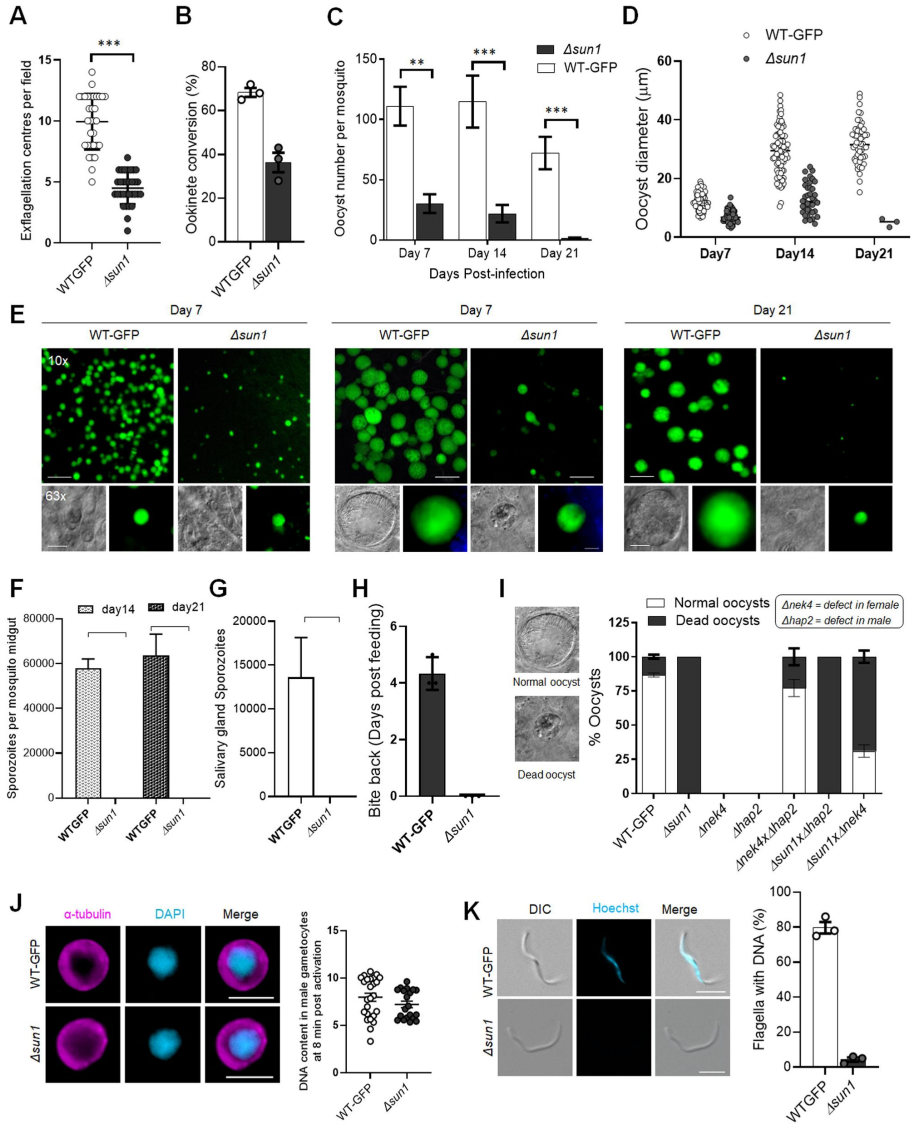


Fig 3.

Deletion of *sun1* affects male gamete formation and blocks parasite transmission.

A. Exflagellation centres per field at 15 min post-activation. n = 3 independent experiments (>10 fields per experiment). Error bar, ± SEM. **B.** Percentage ookinete conversion from zygote. n= 3 independent experiments (> 100 cells). Error bar, ± SEM. **C.** Total number of GFP-positive oocysts per infected mosquito in Δ*sun1* compared to WT-GFP parasites at 7-, 14-and 21-days post infection. Mean ± SEM. n= 3 independent experiments. **D.** The diameter of GFP-positive oocysts in Δ*sun1* compared to WT-GFP parasites at 7-, 14- and 21-days post infection. Mean ± SEM. n= 3 independent experiments. **E.** Mid guts at 10x and 63x magnification showing oocysts of Δ*sun1* and WT-GFP lines at 7-, 14- and 21-days post infection. Scale bar: 50 μm in 10x and 20 μm in 63x. **F.** Total number of midguts sporozoites per infected mosquito in Δ*sun1* compared to WT-GFP parasites at 14- and 21-days post infection. Mean ± SEM. n= 3 independent experiments. **G.** Total number of salivary gland sporozoites per infected mosquito in Δ*sun1* compared to WT-GFP parasites at 21-days post infection. Mean ± SEM. n= 3 independent experiments. **H.** Bite back experiments showing no transmission of Δ*sun1,* while WT-GFP parasites show successful transmission from mosquito to mouse. Mean ± SEM. *n* = 3 independent experiments. **I.** Rescue experiment showing Δ*sun1* phenotype is due to defect in male *sun1* allele. Mean ± SEM. n= 3 independent experiments. **J.** Representative images of male gametocytes at 8 min post activation stained with DAPI and tubulin (left). Fluorometric analyses of DNA content (N) after DAPI nuclear staining (right). The mean DNA content (and SEM) of >30 nuclei per sample are shown. Values are expressed relative to the average fluorescence intensity of 10 haploid ring-stage parasites from the same slide. **K.** Representative images of flagellum (male gamete) stained with Hoechst for DNA (left). The presence or absence of Hoechst fluorescence was scored in at least 30 microgametes per replicate. Mean ± SEM. *n* = 3 independent experiments.

Transcriptomic and lipidomic analysis of Δ*sun1* gametocytes shows no major changes in overall gene expression and lipid metabolism

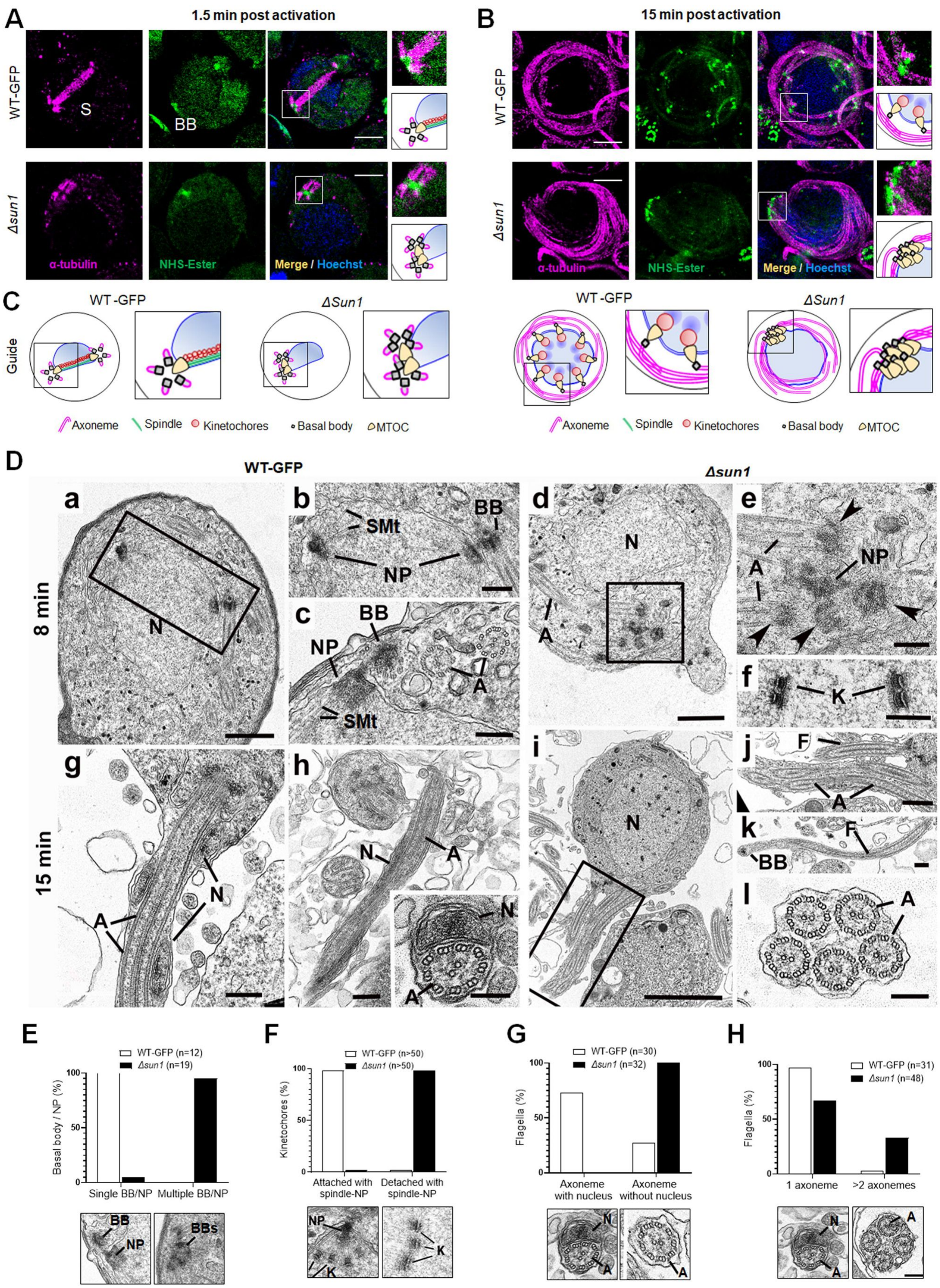
To investigate the effect of *sun1* deletion on transcription, we performed RNA-seq analysis in triplicate on Δ*sun1* gametocytes and in duplicate on WT-GFP gametocytes, at 8LJmin post-activation. Total RNA was extracted to detect changes in gene expression post-activation. A detailed analysis of the mapped reads also confirmed the deletion of *sun1* in the Δ*sun1* strain (Fig. S5D). A relatively small number of differentially expressed genes was identified (Fig. S5E; Supplementary Table S1). Gene ontology (GO)-based enrichment analysis of these genes showed that several upregulated genes coded proteins involved in either lipid metabolic processes, microtubule cytoskeleton organization, or microtubule-based movement (Fig. S5F).

Given the nuclear membrane location of SUN1, we investigated the lipid profile of Δ*sun1* and WT-GFP gametocytes using mass spectrometry-based lipidomic analyses. Lipids are key elements of membrane biogenesis, with most phospholipids derived from fatty acid precursors (Fig. S6A).

We compared the lipid profile of Δ*sun1* and WT-GFP gametocytes, either non-activated (0 min) or activated for 8 min. Both the disruption of *sun1* and the activation of gametocytes caused minimal changes in the amounts of phosphatidic acid (PA), CE, monoacylglycerol (MAG), and TAG. Activated gametocytes had increased levels of PA, TAG, and CE and a decreased level of free fatty acids (FFA), compared to non-activated gametocytes (Fig. S6B, C). Following activation, there was also a differential level of lysophosphatidylcholine (LPC) and of specific fatty acid compositions (Fig S6D). Notably, the Δ*sun1* mutant had altered levels of arachidonic acid (C20:4), a scavenged fatty acid precursor of inflammatory molecules such as leukotrienes and prostaglandins (Fig S6E) (Botte et al., 2013; Dubois et al., 2018), and an increase in myristic acid (C14:0), a product of apicoplast fatty acid synthesis (FASII) (Fig S6E) (Amiar *et al*., 2020; Botte *et al*., 2013).

Ultrastructural Microscopy Reveals Defects in Spindle Formation, Basal Body Segregation and Nuclear Attachment to Axonemes in Δ*sun1* Gametocytes

To define the ultrastructural defects caused by *sun1* deletion, we used ultrastructure expansion microscopy (U-ExM) and transmission electron microscopy (TEM) to examine the morphology of male gametocytes at critical time points -1.5-, 8-, and 15-min post-activation, representing the timings of major transitions in spindle formation, BB segregation, and axoneme elongation (**Fig. 4A-D**, Fig. S7, 8).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WT-GFP
Δsun1
S
BB
α-tubulin
NHS-Ester
Merge / Hoechst
B
15 min post activation
C
Guide
WT-GFP
ΔSun1
Axoneme
Spindle
Kinetochores
Basal body
MTOC
D
WT-GFP
Δsun1
8 min
15 min
SMt
NP
BB
N
A
K
F
E
WT-GFP (n=12)
Δsun1 (n=19)
Basal body / NP (%)
Single BB/NP
Multiple BB/NP
BBs
F
WT-GFP (n>50)
Δsun1 (n>50)
Kinetochores (%)
Attached with spindle-NP
Detached with spindle-NP
G
WT-GFP (n=30)
Δsun1 (n=32)
Flagella (%)
Axoneme with nucleus
Axoneme without nucleus
H
WT-GFP (n=31)
Δsun1 (n=48)
1 axoneme
>2 axonemes

Fig 4.

Ultrastructural analysis of Δ*sun1* gametocytes showing defect in spindle formation and BB segregation

A. Deletion of *sun*1 blocks first spindle formation as observed by ExM of gametocytes activated for 1.5 min. α -tubulin: magenta, amine groups/NHS-ester reactive: green. Basal Bodies: BB; spindle: S. Insets represent the zoomed area marked by the white boxes shown around BB/MTOC highlighted by NHS-ester and tubulin staining. Scale: 5LJµm.

B. ExM images showing defect in BB/MTOC segregation in Δ*sun1* gametocytes activated for 15 min. α -tubulin: magenta, amine groups/NHS-ester reactive: green. Basal Bodies: BB; Insets represent the zoomed area shown around BB/MTOC highlighted by NHS-ester and tubulin staining. More than 30 images were analysed in more than three different experiments. Scale bars:LJ5LJµm.

C. The schematic illustrates structures associated with mitosis and axoneme formation showing the first spindle is not formed, and BB are not separated in Δ*sun1* gametocytes.

D. Electron micrographs of WT-GFP microgametocytes (male) at 8 min (**a-c**) and 15 min (**g, h**) plus Δ*sun1* gametocytes at 8 min (**d-f**) and 15 min (**i-l**). Bars represent 1 µm (**a, d, i**), and 100 nm (**b, c, e, f, g, h, insert, j, k, l**).

a. Low power magnification of WT-GFP microgametocyte showing the central nucleus (N) with two nuclear poles (NP) with a basal body (BB) adjacent to one. The cytoplasm contains several axonemes (A).

b. Enlargement of enclosed area (box) in **a** showing the basal body (BB) adjacent to one nuclear pole (NP).

c. Detail showing the close relationship between the nuclear pole (NP) and the basal body (BB). Note the cross-sectioned axonemes (A) showing the 9+2 microtubular arrangement.

d. Low power magnification of Δ*sun1* cell showing a cluster of electron dense basal structures (enclosed area) in the cytoplasm adjacent to the nucleus (N). A – axonemes.

e. Detail from the cytoplasm (boxed area in **d**) shows a cluster of 4 basal bodies (arrowheads) and portions of axonemes (A) around a central electron dense structure of nuclear pole (NP) material.

f. Detail from a nucleus showing kinetochores (K) with no attached microtubules.

g. Periphery of a flagellating microgamete showing the flagellum and nucleus protruding from the microgametocyte.

h. Detail of a longitudinal section of a microgamete showing the spiral relationship between the axoneme (A) and nucleus (N). **Insert.** Cross section of a microgamete showing 9+2 axoneme and adjacent nucleus (N).

i. Section through a microgametocyte with a central nucleus (N) undergoing exflagellation.

j. Enlargement of the enclosed area (box) in **i** showing one cytoplasmic protrusion containing a single axoneme forming a flagellum (F), while the other has multiple axonemes (A).

k. Longitudinal section through a flagellum (F) with a basal body (B) at the anterior end but note the absence of a nucleus.

l. Cross section showing a cytoplasmic process contain 5 axonemes (A) but no associated nucleus.

E-H. Quantification of Δ*sun1* phenotypes compared to WT-GFP during male gametogenesis. N = Nucleus; BB = Basal Body; NP = Nuclear pole; A = Axonemes.

U-ExM revealed that in WT-GFP gametocytes, spindle microtubules were robustly assembled by 1.5 min post-activation, extending from nuclear MTOCs (spindle poles) to kinetochores. Simultaneously, BB - marked by NHS ester staining - segregated into tetrads, distributed across the cytoplasmic periphery (**Fig. 4A**, S7A). By 8 min, spindles were fully extended, ensuring accurate chromosome segregation, while BB had replicated to nucleate parallel axonemes aligned around the nucleus (Fig. S7B). By 15 min, these processes had culminated in exflagellation, producing mature microgametes containing nuclei tightly associated with BB-derived axonemes (**Fig. 4B**, S7C). In Δ*sun1* gametocytes, these processes were severely disrupted. At 1.5 min, α-tubulin staining showed incomplete or malformed spindles. NHS ester staining revealed BB clumped near one side of the nucleus, indicating a failure in segregation (**Fig. 4A**, S7A). By 8 min, spindles remained rudimentary, and BB segregation was still incomplete. Axoneme elongation proceeded, but BB and nuclear poles failed to align, leading to misconnected or unconnected axonemes (Fig. S7B). At 15 min, gametes had started emerging from cytoplasmic protrusions, but many lacked nuclei due to the absence of proper BB segregation and nuclear compartmentalization (**Fig. 4B**).

We performed TEM analysis of Δ*sun1* and WT-GFP gametocytes at 8- and 15 min post-activation. At 8 min post-activation, many of the wildtype male gametocytes had progressed to the third genome division with multiple nuclear spindle poles within a central nucleus (**Fig. 4Da**; S8A). In many cases, a BB was visible in the cytoplasm closely associated with and connected to the NP (**Fig. 4Db, c**; S8A, B). From the NP, nuclear spindle microtubules (SMt) radiated into the nucleoplasm and connected to kinetochores (**Fig. 4Db, c**; S8A, B). Within the cytoplasm several elongated axonemes, most but not all with the classical 9+2 microtubule arrangement, were visible (**Fig. 4Dc**).

At 8 min in Δ*sun1* gametocytes there were predominately mid stage forms with elongated axonemes running around the nucleus (**Fig. 4Dd**; S8C, D, G). It was also possible to find clumps (groups) of BB (**Fig. 4Dd, e**; S8E, F). Electron densities similar to NPs were observed adjacent to the BB (Fig. 3De; S8H, I, J). An extensive search (100+ sections) failed to identify examples of nuclear spindle formation with attached kinetochores. In contrast to the WT-GFP cells, a significant proportion of Δ*sun1* cells (20%) had groups of centrally located kinetochores with no attached microtubules (naked kinetochores: NK) (**Fig. 4Df**; S8C, D, G). NK were not observed in an extensive search of WT-GFP microgametocytes. (Fig. S8A). Nevertheless, the parallel orientation of the axonemes appeared to be maintained and many displayed the normal 9+2 arrangement (**Fig. 4Dc**).

By 15 min, WT-GFP gametocytes showed examples of exflagellation, with flagella displaying BB at the tip. Cross sections showed gametes with normal axonemes with a 9+2 arrangement and an enclosed nucleus surrounded by a plasma membrane (**Fig. 4Dg, h**). In Δ*sun1* samples, several late stages as well as some mid stages were observed. The late stages had more electron dense cytoplasm (**Fig. 4Di**). The majority had few if any axonemes, but all had a large nucleus with small clumps of electron dense material (**Fig. 4Di**). A few microgametocytes appeared to be undergoing exflagellation with flagellar-like structures protruding from the cell surface, often with multiple axonemes (**Fig. 4Dj, l**). However, they were abnormal, lacking a nucleus and often with variable numbers (1 to 6) of normal 9+2 axonemes (**Fig. 4Dj, k, l**). There appeared to be a disconnect between the axonemes and the nucleus during exflagellation. There was evidence of chromatin condensation to form microgamete nuclei but no connection between the nucleus and axoneme during exflagellation, resulting in microgametes lacking a nucleus. This Δ*sun1* mutant appeared to have a problem with separation of NPs and nuclear spindle formation, resulting in no genome segregation. The nuclear poles were difficult to find and did not appear to have been able to divide or move apart (**Fig. 4Dd, e**; S8E, F, H, I, J). It is possible that this lack of spindle pole division and movement was responsible for the BB remaining clumped together in the cytoplasm. It is also possible that clumping of the BB may explain the presence of multiple axonemes associated with the exflagellated structures. The quantification data for the Δ*sun1* phenotypes are shown in **Fig. 4E-H** and Fig. S8K.

Together, these findings highlight the pivotal role of SUN1 in coordinating the nuclear and cytoplasmic compartments during male gametogenesis. Its absence disrupts spindle formation, BB segregation, and the physical attachment of axonemes to the nucleus, resulting in anucleate microgametes incapable of fertilization.

SUN1 Interactome Reveals Associations with Nuclear Envelope, ER and Chromatin Components

To identify protein interaction partners of PbSUN1, we performed GFP-based immunoprecipitation (IP) using a nanobody targeting SUN1-GFP in lysates of purified gametocytes activated for 6 to 8 min. This time point was chosen as it coincides with peak nuclear expansion and axoneme formation. To stabilize transient or weak interactions, protein cross-linking with 1% paraformaldehyde was used prior to IP. Co-immunoprecipitated proteins were identified using LC-MS/MS of tryptic peptides and were analysed using principal component analysis for duplicate WT-GFP and SUN1-GFP precipitations. (**Fig. 5A**, Supplementary Table S2).

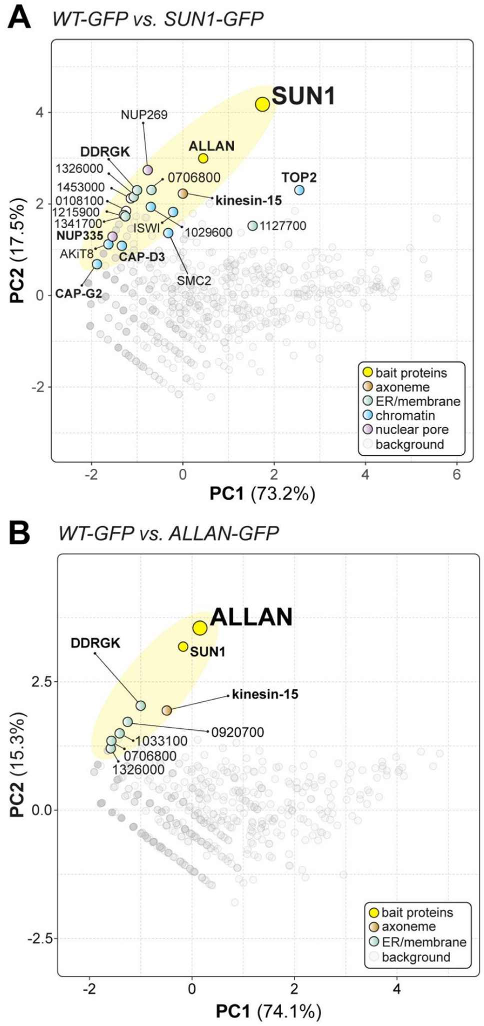
Co-variation with SUN1 was found for proteins of the nuclear envelope, chromatin and ER-related membranes. Notably, SUN1 co-purified with nuclear pore proteins (NUP269, NUP335), membrane proteins, including a likely ER component (DDRGK-domain containing UFM1 E3 ligase, PBANKA_0927700), chromatin-related factors (e.g. condensin I/II subunits, topoisomerase II and the kinetochore subunit AKit-8), and the cytoplasmic and male gametocyte-specific kinesin-15 (PBANKA_145880). This suggests a role for SUN1 in bridging chromatin (through condensin II) on the nuclear side and the cytoskeleton (through kinesin-15) on the cytoplasmic side of the nuclear envelope, potentially in association with nuclear pores (**Fig. 5A**) in a similar fashion to what has been proposed in the plant Arabidopsis (Ito et al., 2024; Sakamoto et al., 2022).

PbSUN1 also interacted with proteins harbouring a divergent carbohydrate-binding domain (like SUN1 itself), such as the allantoicase-like protein ALCC1 (Sayers et al. 2024), here referred to as ALLAN (PBANKA_1144200) and PBANKA_0209200, an ER-Golgi protein with a mannose-binding domain (**Fig. 5A**, Supplementary Table S2). ALLAN is an uncharacterized orthologue of allantoicase, an enzyme in purine metabolism. We could detect no KASH-like or lamin-like proteins in the co-immunoprecipitates. To further explore the interaction between ALLAN and SUN1, we performed the reciprocal ALLAN-GFP, from lysates of gametocytes 6-min post activation. Amongst the interactors we found SUN1 and its interactors, DDRGK-domain containing protein and kinesin-15 (**Fig. 5B**, Supplementary Table S2). The results from the immunoprecipitation experiments suggest that *Plasmodium* SUN1 functions in a non-canonical fashion via an interaction with ALLAN to tether chromatin to the nuclear envelope and possibly to the cytoplasmic cytoskeleton via kinesin-15 (**Fig. 5C**).

We hypothesized that SUN1 is at the centre of multiple molecular interactions during male gametogenesis, coordinating NE remodelling, spindle organization, and BB/axoneme attachment, with ALLAN as a key interactor near the nuclear MTOC.

ALLAN is Located at Nuclear Poles (NP) and influences Basal Body/NP Segregation in Male Gametogenesis

To reveal ALLAN's function in the *Plasmodium* life cycle, we generated a C-terminal GFP-tagged ALLAN parasite line, confirming correct integration by PCR (Fig. S9A, B) and detecting a protein of ~85 kDa by Western blot (Fig. S9C). ALLAN-GFP parasites displayed normal growth and completed



C

PBANKA	annotation	AA	GFP 1	GFP 2	SUN1 1	SUN1 2	ALLAN 1	ALLAN 2
1430900	**SUN domain-containing protein**	**872**	**-**	**-**	**47**	**49**	**29**	**2**
1304400	**allantoicase ALLAN**	**478**	**-**	**-**	**19**	**10**	**17**	**14**
1453000	conserved Plasmodium protein	1145	-	-	15	-	-	-
0108100	conserved Plasmodium protein	1109	-	-	10	-	-	-
1341700	ARM domain, unknown function	2004	-	-	9	-	-	-
1458800	**kinesin-15**	**1414**	**-**	**-**	**16**	**2**	**14**	**-**
1454600	NUP269	2260	-	-	28	-	-	-
0807900	NUP335	2840	-	-	5	-	-	-
0927700	**DDRGK domain - UFM1 E3 ligase**	**1889**	**-**	**-**	**17**	**-**	**5**	**1**
1127800	DnaJ protein (chaperone)	379	2	4	16	5	-	-
0706800	DnaJ protein (chaperone)	646	-	-	14	-	3	-
1326000	atlasin-like, GTP-binding protein	1297	-	-	15	-	2	-
1215900	dynamin-like	1262	-	-	9	-	-	-
0920700	fam-b protein	439	-	-	7	-	6	-
1011400	**DNA topoisomerase 2 (TOP2)**	**1472**	**3**	**4**	**35**	**13**	**4**	-
0712100	DNA helicase ISWI	1018	-	-	10	2	-	-
1416900	SMC2 (condensin)	1227	-	-	7	2	-	-
1029600	Myb domain protein	376	-	-	8	2	-	-
0301800	condensin-2 complex subunit D3	1706	-	-	4	-	-	-
0213200	kinetochore protein AKiT8	2347	-	-	3	-	-	-
1312000	condensin-2 complex subunit G2	1181	-	-	3	-	-	-

Fig. 5.

Reciprocal co-immunoprecipitation of PbSUN1-GFP and ALLAN-GFP during male gametogony

A. Projection of the first two components of a principal component analysis (PCA) of unique peptides derived from two SUN1-GFP (and WT-GFP) immunoprecipitations with GFP-trap (peptide values: Supplementary Table S2). A subset of proteins is highlighted on the map based on relevant functional categories. **B.** Similar to panel A, but now for the allantoicase-like protein ALLAN (PBANKA_1304400). **C.** Selected proteins, their size and corresponding gene ID and representation by the number of peptides in either WT-GFP, PbSUN1-GFP or ALLAN-GFP precipitates.

the life cycle, indicating that GFP tagging did not impair protein function.

During asexual blood stages, ALLAN-GFP exhibited a diffuse nucleoplasmic signal in trophozoites and schizonts with distinct focal points adjacent to dividing DNA, consistent with a role in mitotic regulation (Fig. S9D). However, by late schizogony, the ALLAN signal had diminished. In male and female gametocytes (Fig. S9E) and during zygote to ookinete transformation (Fig. S9F), ALLAN-GFP showed a spherical distribution around Hoechst-stained nuclear DNA and was enriched to form distinct focal points. During oocyst development and liver schizogony, ALLAN-GFP also exhibited distinct focal points adjacent to dividing DNA (Fig. S9G, H) suggesting a role in mitotic regulation, like in asexual blood stages (Fig. S9D).

In activated male gametocytes, ALLAN-GFP rapidly localized within a min post-activation to the NE, forming strong focal points that correlated with spindle poles or MTOCs (**Fig. 6A**, Supplementary Video S2). This localization persisted as ploidy increased from 1N to 8N in successive rounds of genome replication (**Fig. 6A**). Using U-ExM, ALLAN-GFP was resolved at the INM, enriched at spindle poles marked by NHS-ester staining, but absent from BB, where axonemes form (**Fig. 6B, C**). The location of ALLAN-GFP showed no overlap with that of kinesin-8B, a BB marker, or that of kinetochore marker NDC80 (**Fig. 6D, F**). Conversely, crosses with EB1-mCherry, a spindle marker, revealed an overlap at spindle poles (**Fig. 6E**) suggesting a role for ALLAN in nuclear spindle pole organization. Thus, the spatial relationship of SUN1 and ALLAN suggests coordinated roles in nuclear architecture and division of labour. SUN1 likely spans the NE, linking nuclear and cytoplasmic compartments, while ALLAN localizes more specifically to nuclear MTOCs.

The proximity of both SUN1 and ALLAN at spindle poles is consistent with their collaboration to align spindle microtubules with nuclear and cytoplasmic MTOCs and ensure that chromosomal segregation aligns with axoneme formation. We performed functional studies of ALLAN deletion mutants to examine its contribution to spindle dynamics and its potential as a key player in rapid *Plasmodium* closed mitosis.

Functional Role of ALLAN in Male Gametogenesis

We generated Δ*allan* mutants using double-crossover homologous recombination. Correct integration and successful gene deletion were confirmed via PCR (Fig. S10A, S10B) and qRT-PCR, with no residual *allan* transcription in Δ*allan* lines (Fig. S10C). There was no phenotype in asexual blood stages, but Δ*allan* mutants exhibited significant defects in mosquito stages.

Δ*allan* parasites showed no marked reduction in male gamete exflagellation compared to WT-GFP controls (**Fig. 7A**), and ookinete conversion rates were only slightly reduced (**Fig. 7B**). However, oocyst counts were significantly lower in Δ*allan*-parasite infected mosquitoes (**Fig 7C**), with diminished oocyst size (Fig S10D) and significant decrease in sporozoite number at 14-and 21-days post-infection (**Fig. 7D**). Though salivary gland sporozoites were significantly decreased (Fig S10E), Δ*allan* parasites were successfully transmitted to mice in bite-back experiments, showing that some viable sporozoites were produced (**Fig. 7E**).

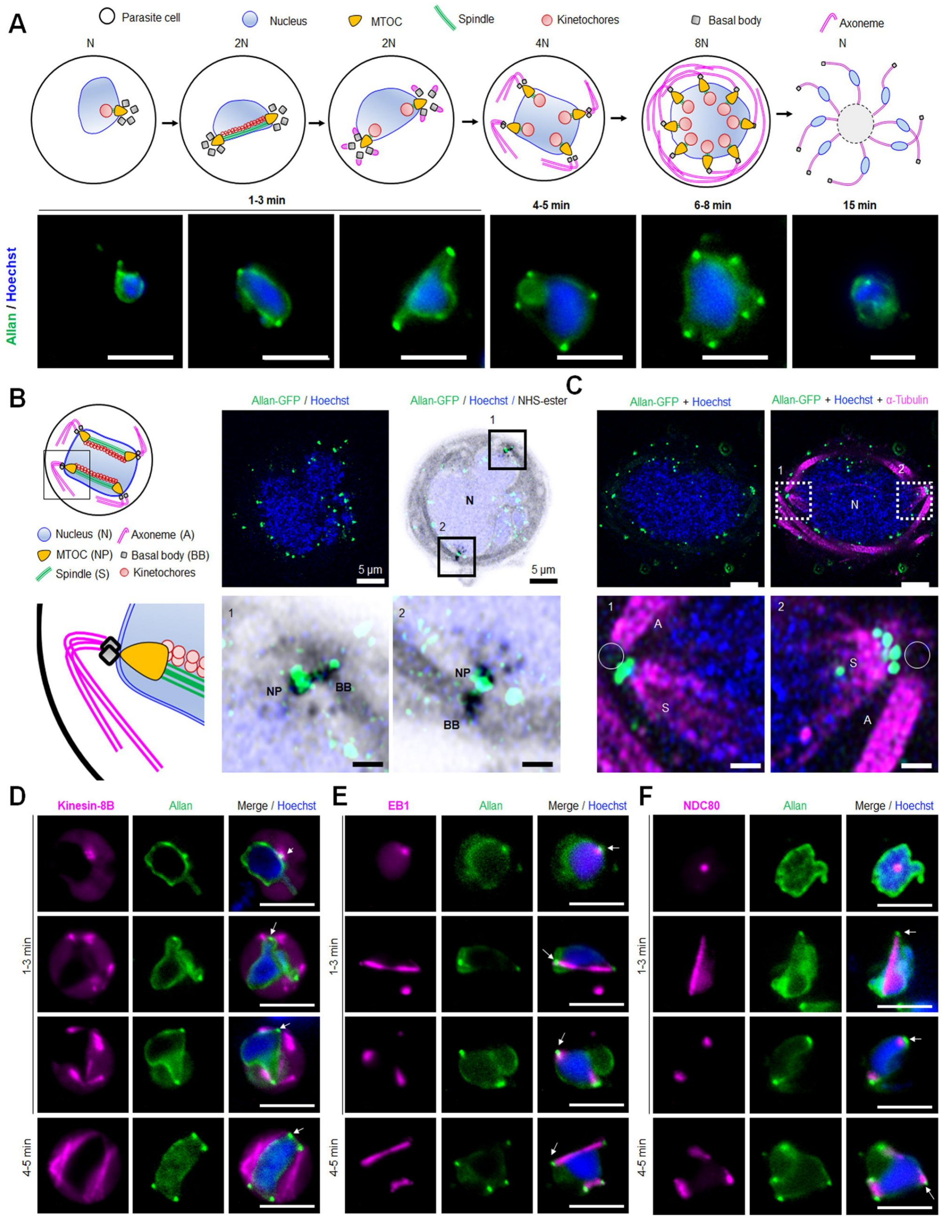


Fig. 6.

Location of ALLAN-GFP during male gametogenesis

A. The schematic on the upper panel illustrates the process of male gametogenesis. N, ploidy of nucleus. Live cell images showing the location of ALLAN-GFP (green) at different time points(1-15min) during male gametogenesis. Representative images of more than 50 cells with more than three biological replicates. Scale bar: 5 µm. **B.** ExM images showing location of ALLAN-GFP (green) detected by anti-GFP antibody compared to nuclear pole (NP)/MTOC and BB stained with NHS ester (grey) in gametocytes activated for 8 min. Scale bar: 5 µm. Representative images of more than 20 cells from two biological replicates. Insets represent the zoomed area shown around NP/MTOC and BB highlighted by NHS-ester. Scale bar: 1 µm. **C.** ExM images showing the location of ALLAN-GFP (green) compared to spindle and axonemes (magenta) detected by anti-GFP and antI-tubulin staining respectively in gametocytes activated for 8 min. Representative images of more than 20 cells from two biological replicates. Scale: 5 µm. Insets represent the zoomed area shown around spindle /axonemes highlighted by tubulin and GFP staining. Basal Bodies: BB; Spindle: S; Axonemes: A; Nucleus: N. Scale bar: 1 µm. **D, E, F.** Live cell imaging showing location of ALLAN-GFP (green) in relation to the BB and axoneme marker, kinesin-8B-mCherry (magenta) (D); spindle marker, EB1-mCherry (magenta) (E); and kinetochore marker, NDC80-mCherry (magenta) (F) during first mitotic division (1-3 min) of male gametogenesis. Arrows indicate the focal points of ALLAN-GFP. Representative images of more than 20 cells with three biological replicates. Scale bar: 5LJµm.

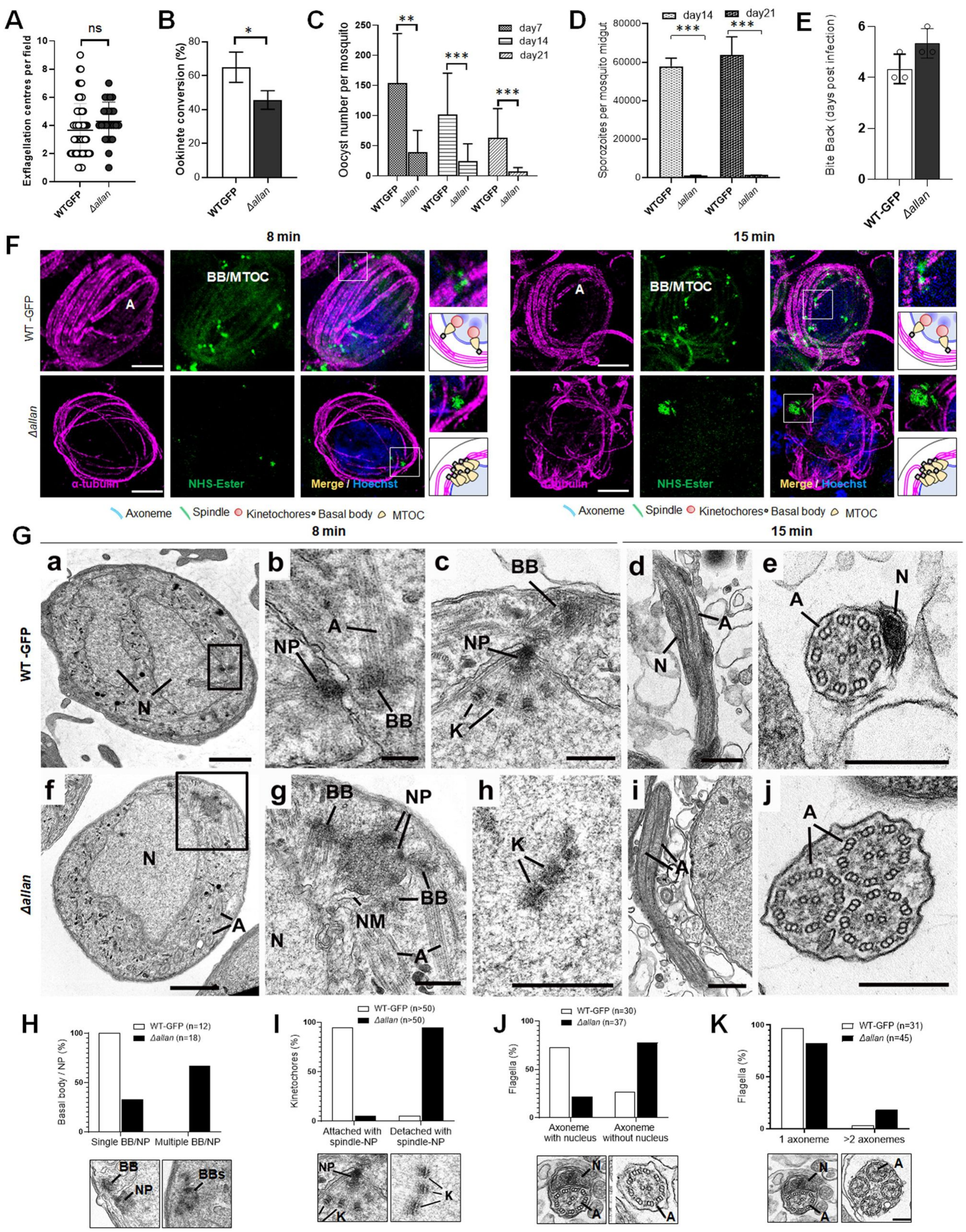

A
B
C
D
E
F
G
H
I
J
K
Exflagellation centres per field
Ookinete conversion (%)
Oocyst number per mosquito
Sporozoites per mosquito midgut
Bite Back (days post infection)
WTGFP
Δallan
WT-GFP
day7
day14
day21
8 min
15 min
BB/MTOC
α-tubulin
NHS-Ester
Merge / Hoechst
Axoneme
Spindle
Kinetochores
Basal body
MTOC
Basal body / NP (%)
Single BB/NP
Multiple BB/NP
Kinetochores (%)
Attached with spindle-NP
Detached with spindle-NP
Flagella (%)
Axoneme with nucleus
Axoneme without nucleus
1 axoneme
>2 axonemes
WT-GFP (n=12)
Δallan (n=18)
WT-GFP (n>50)
Δallan (n>50)
WT-GFP (n=30)
Δallan (n=37)
WT-GFP (n=31)
Δallan (n=45)

Fig. 7.

Deletion of ALLAN impairs male gametogenesis by blocking BB segregation

A. Exflagellation centres per field at 15 min post-activation in Δ*allan* compared to WT-GFP parasites. n≥3 independent experiments (>10 fields per experiment). Error bar, ± SEM. **B.** Percentage ookinete conversion from zygote. n≥3 independent experiments (> 100 cells). Error bar, ± SEM. **C.** Total number of GFP-positive oocysts per infected mosquito in Δ*allan* compared to WT-GFP parasites at 7-, 14-and 21-days post infection. Mean ± SEM. n≥3 independent experiments. **D.** Total number of sporozoites in oocysts of Δ*allan* compared to WT-GFP parasites at 14- and 21-days post infection. Mean ± SEM. n≥3 independent experiments. **E.** Bite back experiments reveal successful transmission of Δ*allan* and WT-GFP parasites from mosquito to mouse. Mean ± SEM. *n* = 3 independent experiments. **F.** ExM images of gametocytes activated for 8- and 15-min showing MTOC/BB stained with NHS ester (green) and axonemes stained with anti-tubulin antibody (magenta). Axonemes: A; Basal Bodies: BB; Microtubule organising centre: MTOC. Insets represent the zoomed area shown around BB/MTOC highlighted by NHS-ester and tubulin staining. More than 30 images were analysed in more than three different experiments. Scale bars:LJ5LJµm. **G.** Electron micrographs of WT-GFP microgametocytes at 8 mins (a-c) and 15 mins (d, e) and the Δ*allan* at 8 min (f-h) and 15 min (I, j). Bars represent 1 µm (a, f) and 200 nm in all other images.

a. Low power image of a microgametocyte showing the nucleus (N) with two NP complexes (arrows) consisting of the basal body, NP and attached kinetochores. Axonemes (A) are present in the cytoplasm.

b. Enlargement showing the nuclear pole (NP), associated basal body (BB) and axonemes (A).

c. Detail of the nuclear pole (NP) showing kinetochores (K) attached the spindle microtubules. Note the basal body (BB) adjacent to the nuclear pole (NP).

d. Longitudinal section of a microgamete showing the nucleus (N) closely associated with the axonemes (A).

e. Cross section through a microgamete showing the nucleus (N) and axoneme (A) enclosed plasma membrane.

f. Lower magnification of a microgametocyte showing a nucleus (N) with an adjacent clump of electron dense structures (enclosed area) and axonemes (A) in the cytoplasm.

g. Enlargement of the enclosed area in f showing multiple basal bodies (BB) and unseparated nuclear poles (NP) enclosed by portions of nuclear membrane (NM). N – nucleus.

h. Detail from a nucleus showing several kinetochores (K) with no associated spindle microtubules.

i. Longitudinal section of an exflagellating cytoplasmic process consisting of two axonemes (A) but no nucleus.

j. Cross section through an exflagellating cytoplasmic process showing the presence of multiple axonemes (A) but the absence of any nucleus.

H to K. Quantification of Δ*allan* phenotype compared to WT-GFP during male gametogenesis. N = Nucleus; BB = Basal Body; NP = Nuclear pole; A = Axonemes.

U-ExM analysis revealed a striking defect in Δ*allan* male gametocyte morphology. At 8 min post-activation, NHS-ester staining indicated clustered BB, with incomplete segregation and misalignment relative to the nuclear MTOCs (**Fig. 7F**). Despite normal axoneme elongation, spindle organization was disrupted, as seen in nonsegregated BB/MTOCs (**Fig. 7F**).

To further explore the structural changes, we used electron microscopy analysis of gametocytes fixed at 8 min and 15 min post activation (**Fig. 7G**). At 8 min in the WT parasite there was a large lobated nucleus exhibiting multiple NPs. Radiating from the NPs were microtubules forming the nuclear spindle with associated kinetochores. BB was closely associated with the NP on the cytoplasmic side from which extended an axoneme (**Fig. 7Ga, b, c**). By 15 min, exflagellation had occurred and free microgametes were observed consisting of a 9+2 axoneme with closely associate nucleus enclosed by a unit membrane (plasmalemma) (**Fig. 7Gd, e**).

In the Δ*allan* parasites at 8 min, numerous mid stage microgametocytes were observed. However, in contrast to the WT parasites, the BB appeared to be clumped together although axoneme formation was similar to that in WT. (**Fig. 7Gf, g**). NP-like structures lacking spindle microtubules were often associated with clumps of electron dense material (**Fig. 7Gg**; S11D, E). Examination of sections of mid stage gametocytes (50+), revealed several cells (~20%) with free kinetochores and no associated spindle microtubules (Fig. 6Gh; S11A, B, C). It was difficult to find individual NPs or obvious nuclear spindles although very rare examples were observed (>2%).

By 15 min post-activation in Δ*allan* gametocytes, several cells had exflagellated with the formation of numerous free gametes – the majority with no associated nucleus (**Fig. 7Gi**; S11G, H). Some had multiple axonemes (**Fig. 7Gi, j**; S11H) but there were many fewer examples than seen in the Δ*sun1* mutant. However, unlike the Δ*sun1* line, a few male gametes with axonemes and associated nuclei were observed. The Δ*allan* phenotype quantitative data are shown in **Fig. 7H-K** and Fig. S11I.

These Δ*allan* defects suggest that ALLAN is important for the separation of the NPs with formation of spindle microtubules and kinetochore attachment. The low incidence of spindle assembly and impaired NP organization in Δ*allan* gametocytes is reminiscent of the Δ*sun1* mutant phenotype, reinforcing the relationship between these two proteins. The presence of a few normal nuclear spindle poles with attached kinetochores and formation of a few microgametes with axoneme and nucleus may explain the transmission observed for this mutant.

Evolution of a Novel SUN1-ALLAN Axis in Haemosporida

The non-canonical SUN1-ALLAN complex in *P. berghei* is a unique apicomplexan adaptation, where NE and MTOC dynamics diverge from those of other eukaryotes. To trace the evolution of this interaction, we examined SUN1, ALLAN, and potential interactors (e.g., KASH proteins and lamins) across alveolates (apicomplexans, dinoflagellates and ciliates) and model organisms such as yeast, humans, and *Arabidopsis* (**Fig. 8**, Supplementary Table S3).

SUN domain proteins generally fall into two families: C-terminal SUN domain and N-terminal/mid SUN domain proteins (Graumann et al., 2014). Most eukaryotes have representatives of both families. *P. berghei* has two SUN domain proteins (Kandelis-Shalev et al, 2024), one of each family with PbSUN1 having a C-terminal SUN domain. Allantoicase-like proteins often consist of two nearly identical domains, belonging to the galactose-binding-like family that are remarkably similar to the SUN domain, (**Fig. 8A**). Among apicomplexans, two allantoicase subtypes are present due to duplication in the ancestor of all Apicomplexa: AKiT8 (Brusini et al., 2022) a kinetochore-associated protein, and ALLAN, the allantoicase-like subtype. Notably, *Plasmodium* is one of the few apicomplexans retaining both subtypes (i.e. only AKiT is retained in Toxoplasma). Extreme sequence divergence complicates phylogenetic classification of allantoicase-like proteins among apicomplexans, but length-based classification distinguished the longer AKiT8 (~2300 aa) from ALLAN-like proteins (~800 aa; **Fig. 8B**). Despite using previously developed Hidden Markov Model and structural searches (Benz et al., 2024), we found no evidence of KASH-domain proteins or lamins in amongst apicomplexans, similar to what was reported in prior studies (**Fig. 8B**) (Koreny and Field, 2016).

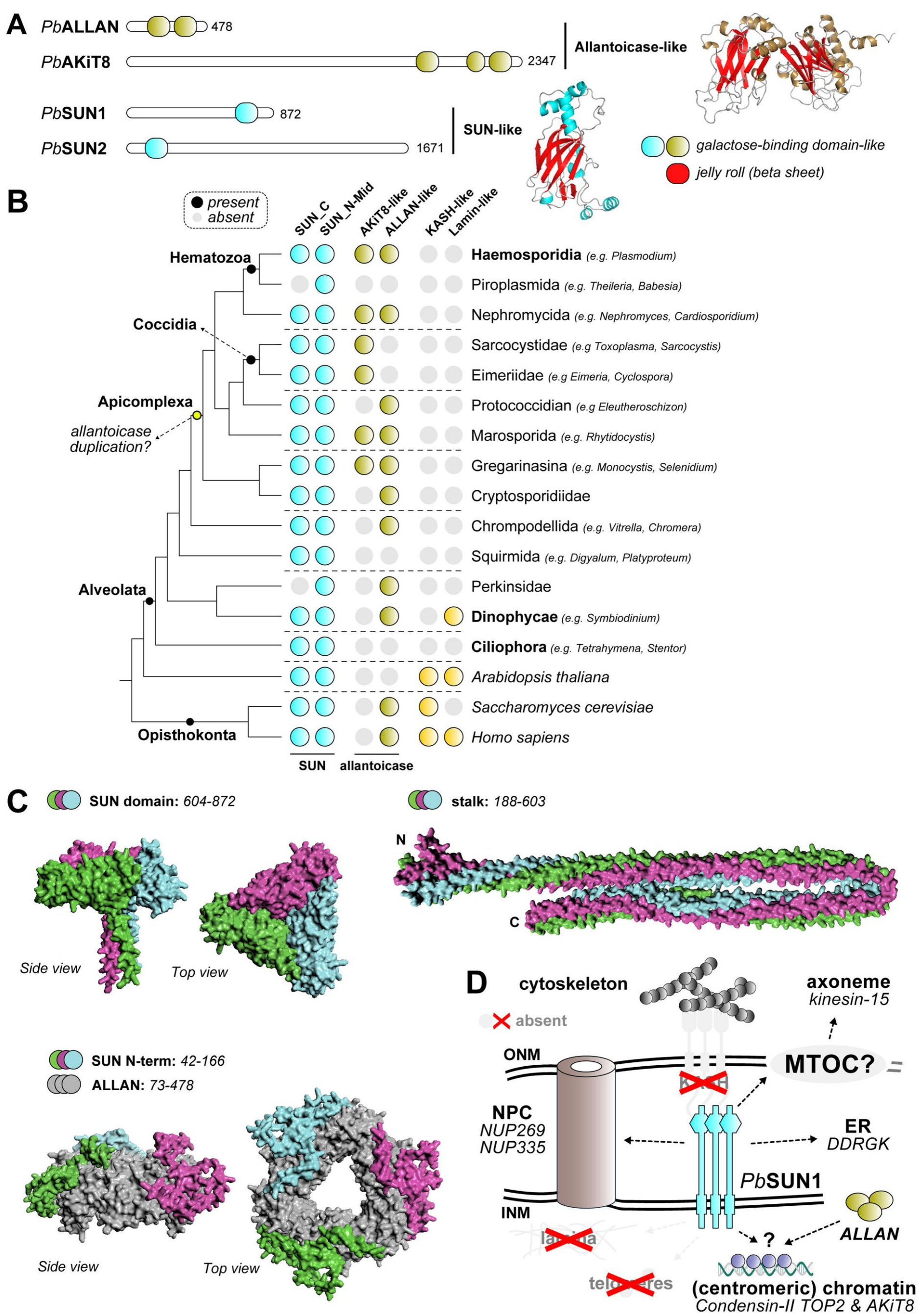


Fig 8.

Evolution and Structure of the SUN1-ALLAN interaction

A. Domain analysis shows two proteins with allantoicase domain and two proteins with SUN-domain in *P. berghei.* The SUN domain and two domains comprising allantoicases are part of the same galactose-binding domain family, with a strikingly similar fold. **B.** Phylogenetic profiles showing the presence of SUN-, ALLAN- and KASH-domain and lamin proteins in Apicomplexa and a selection of other eukaryotes, including two model species *Homo sapiens* and *Arabidopsis thaliana.* **C.** AlphaFold3-modelled interaction between ALLAN and SUN1 based on separate domains (no full structure could be modelled). The SUN1 C-terminus forms a trimeric complex (pTM:0.37) similar to a trimeric ALLAN complex (grey) with the N-terminus of SUN1 interacting with ALLAN (pTM:0.55). This N-terminal domain is unique to Haemosporida. **D.** Overview in similar style as **Fig. 1** of main interactors for putative localization at the nuclear envelope for ALLAN and SUN1 during male gametogenesis. Structures in grey have not been found to be associated with SUN1.

AlphaFold3 (AF3) modelling suggested that SUN1 and ALLAN interact specifically in Haemosporida through the N-terminal domain of SUN1, a region that is absent from other apicomplexan and/or other eukaryotic SUN1 family proteins (**Fig. 8C**). We could not detect similar interactions between SUN1-like and ALLAN-like orthologs amongst other apicomplexans, ruling out a rapidly evolving interaction. Based on these structural predictions and the super-resolution imaging of SUN1-and ALLAN-GFP lines (**Fig. 2D, E**; **Fig. 6B, C**), we propose that ALLAN resides on the nucleoplasmic side of the NE, while SUN1's C-terminal domain likely extends toward the ONM to interact with proteins involved in axoneme and cytoskeletal dynamics in the nucleoplasm (**Fig. 8D**). This specificity may underscore the restricted nature of a SUN1-ALLAN complex to Haemosporida, and thus specific adaption of the allantoicase-like protein ALLAN for nuclear envelope dynamics.

Discussion

This study identifies the SUN1-ALLAN complex as a novel and essential mediator of NE remodelling and bipartite MTOC coordination during *P. berghei* male gametogenesis. *Plasmodium* lacks KASH-domain proteins and lamins, and a canonical LINC complex, relying on a highly divergent mechanism to tether nuclear and cytoplasmic compartments during rapid closed mitosis (Rout et al., 2017).

Our results reveal that SUN1 and ALLAN form a unique complex essential for spindle assembly, BB segregation, and axoneme organization, maintaining an intact NE throughout. Disruption of either protein causes spindle and MTOC miscoordination, leading to defective male gametes incapable of fertilization. Additionally, SUN1 deletion alters lipid homeostasis and NE dynamics, underscoring its multifaceted role in nuclear organization and metabolism. These findings establish the SUN1-ALLAN axis as a crucial evolutionary innovation in *Plasmodium*.

In *P. berghei* male gametogenesis, mitosis is closed and extraordinarily rapid with features that diverge significantly from those of classical models. Within just 8 min, there are three rounds of DNA replication (1N to 8N), accompanied by the assembly of axonemes on cytoplasmic BB (Guttery *et al*., 2022). KASH-domain proteins and lamins are absent, yet *Plasmodium* achieves precise nuclear and cytoskeletal coordination. Our findings reveal that this is accomplished through a non-canonical SUN1-ALLAN complex. SUN1 located at the NE, forms loops and folds to accommodate rapid nuclear expansion, and ALLAN facilitate spindle pole coordination.

Recent studies on SUN1-domain containing proteins have been reported in some Apicomplexa parasites including *Toxoplasma* and *Plasmodium* (Kandelis-Shalev et al., 2024; Sayers et al., 2024; Wagner et al., 2023). SUN-like protein-1 of *Toxoplasma gondii* shows a mitotic spindle pole (MTOC) location and is important for nuclear division (Wagner *et al*., 2023). *Plasmodium spp.* encode two SUN proteins (SUN1 and SUN2), that show a nuclear membrane location and play roles in nuclear division and DNA repair during the blood stage of *P. falciparum* (Kandelis-Shalev *et al*., 2024). Recent work by Sayers et al. (2024) demonstrated that PbSUN1-HA was associated with the NE and suggested that it may be required to capture the intranuclear MTOC onto the NE (Sayers *et al*., 2024). They also showed that it forms a complex with ALCC1 (the protein we have designated ALLAN). They showed that SUN1 is essential for fertile male gamete formation and that in SUN1-KO lines basal bodies are aggregated. Our study extends their findings by providing live-cell imaging of SUN1 and ALLAN and their coordination with key mitotic markers such as NDC80, EB1, and Kinesin-8B. We generated and characterised both SUN1 and ALLAN knockout parasites, performing ultrastructural analyses using TEM and extensive functional studies. Consistent with the results of Sayers et al., we observed basal body clumping in SUN1-KO, and showed that kinetochore attachment to the spindle is compromised in both mutant parasites. Our work emphasizes the role of the nuclear envelope in the rapid cell division during male gametogenesis. Collectively, the findings from both studies highlight the importance of the SUN1–ALLAN complex in Plasmodium cell division (Sayers *et al*., 2024).

Plasmodium SUN1 shares some similarities with SUN proteins in other organisms like *S. cerevisiae and Schizosaccharomyces pombe* (Fan et al., 2022; Hagan and Yanagida, 1995), but its interaction with ALLAN is a highly specialized adaptation in *Plasmodium*. The location of SUN1 at the nuclear MTOC, distinct from axonemal markers like kinesin-8B, underscores its role in nuclear compartmentalization. Similarly, the absence of functional spindle formation and kinetochore attachment in Δ*sun1* mutants is reminiscent of phenotypes seen in mutants of spindle-associated proteins in *Plasmodium*, such as EB1, ARK2 and kinesin-8X (Zeeshan *et al*., 2023; Zeeshan et al., 2019b). However, SUN1's role extends specifically to NE remodelling and the organization of the inner acentriolar MTOC.

Our study demonstrates that SUN1 is primarily located in the space between the INM and ONM, with the NE but facing the nucleoplasm via it's N-terminus, forming dynamic loops and folds to accommodate the rapid expansion of the nucleus during three rounds of genome replication. These loops serve as structural hubs for spindle assembly and kinetochore attachment at the nuclear MTOC, separating nuclear and cytoplasmic compartments. The absence of SUN1 disrupts spindle formation, causing defects in chromosome segregation and kinetochore attachment, while BB segregation remains incomplete, resulting in clumped basal bodies and anuclear flagellated gametes. This phenotype highlights the role of SUN1 in maintaining the nuclear compartment of the bipartite MTOC.

ALLAN, a novel allantoicase-like protein, has a location complementary to that of SUN1, forming focal points at the inner side of the NE. Like SUN1, ALLAN is also not essential during asexual erythrocytic stage proliferation. Functional studies following ALLAN gene deletion revealed similar defects in MTOC organization, with impaired spindle formation, kinetochore attachment, and nuclear-cytoplasmic coordination during flagellated gamete assembly. These findings indicate that SUN1 and ALLAN work together, replacing the role of lamins and KASH proteins in coordinating nuclear and cytoskeletal dynamics.

Interestingly, our interactome analysis identified kinesin-15 as a putative interactor of SUN1 and ALLAN, suggesting a possible link between this motor protein and nuclear remodelling. Previous studies have shown that kinesin-15 is essential for male gamete formation and is located partially at the plasma membrane and nuclear periphery, albeit mostly cytoplasmic (Zeeshan et al., 2022b). On the nuclear side, we found subunits of the condensin-II complex, similar to what was found in Arabidopsis (Ito *et al*., 2024; Sakamoto *et al*., 2022). These interactions with SUN1 and ALLAN raise the possibility of a broader network of proteins facilitating nuclear and cytoskeletal coordination during male gametogenesis.

Transcriptomic analysis of the Δ*sun1* mutant showed upregulation of genes involved in lipid metabolism and microtubule organization. Lipidomic profiling further confirmed substantial alterations in the lipid composition of SUN1-deficient gametocytes. Notably, PA and CE, both critical for membrane curvature and expansion, were reduced in the Δ*sun1* mutant, whereas MAG and DAG levels were elevated. These disruptions likely reflect impaired NE expansion and structural integrity during gametogenesis. Additionally, the altered levels of specific fatty acids, such as arachidonic acid and myristic acid, suggest perturbations in both scavenged and apicoplast-derived lipid pathways..

The SUN1-ALLAN complex exemplifies a lineage-specific adaptation in *P. berghei*, highlighting the evolutionary plasticity of NE remodelling mechanisms across eukaryotes. SUN-domain proteins, a hallmark of LINC complexes, are widely conserved and diversified across eukaryotic lineages, yet *Plasmodium* and related apicomplexans lack key canonical LINC components.

Our analyses suggest that the SUN1-ALLAN complex is specific to Haemosporida. While SUN-domain proteins are conserved across most eukaryotes, the allantoicase-like protein ALLAN is a novel addition to the nuclear architecture of apicomplexans. Phylogenetic profiling indicates that ALLAN likely arose from a duplication of an allantoicase in an early apicomplexan ancestor, likely the ancestor to all coccidians and hematozoa. This event gave rise to two subtypes: AKiT8, associated with kinetochores (Brusini *et al*., 2022), and ALLAN, specialized in NE dynamics.

The absence of KASH proteins and lamins in apicomplexans raises fascinating questions about how NE and cytoskeletal coordination evolved in this group. Comparative studies in ichthyosporeans, close relatives of fungi and animals, have identified intermediate forms of mitosis involving partial NE disassembly (Shah et al., 2024). In contrast, *Plasmodium* employs a highly streamlined closed mitosis, likely driven by the requirement for rapid nuclear and cytoskeletal coordination during male gametogenesis. The discovery of the SUN1-ALLAN axis suggests that apicomplexans have developed alternative strategies to adapt the NE for their cellular and mitotic requirements. These findings raise intriguing questions about the evolutionary pressures that shaped this unique nuclear-cytoskeletal interface in apicomplexans. The presence of similar or other non-canonical complexes in other organisms remains an open question, with potential implications for understanding the diversity of mitotic processes.

This study opens several avenues for future research into the biology of the malaria parasite and its unique adaptations for mitosis. While the SUN1-ALLAN complex has been shown to play a central role in bipartite MTOC organisation during male gametogenesis, many questions remain about its broader role and potential applications. Another key question concerns the evolutionary origins of the SUN1-ALLAN axis; comparative studies across apicomplexans, dinoflagellates, ciliates, and other non-model eukaryotes may help trace the evolution of this complex. The presence of similar non-canonical complexes in other rapidly dividing eukaryotic systems may shed light on the diversity of mitotic adaptations beyond *Plasmodium*. A functional further exploration of associated proteins, such as kinesin-15, may help elucidate how the SUN1-ALLAN complex integrates with other cellular machinery and structures. Investigating whether kinesin-15, a motor protein, interacts directly with SUN1 or ALLAN and has a catalytic role in nuclear-cytoskeletal remodelling may be critical to further our understand of how the system achieves such rapid mitotic coordination.

Methods

Ethics statement and mice details

The animal work passed an ethical review process and was approved by the United Kingdom Home Office. Work was carried out under UK Home Office Project Licenses (PDD2D5182 and PP3589958) in accordance with the UK 'Animals (Scientific Procedures) Act 1986'. Six-to eight-week-old female CD1 outbred mice from Charles River laboratories were used for all experiments. The mice were maintained under a 12LJh light and 12LJh dark (7 am till 7LJpm) cycle, at a temperature between 20 and 24LJ°C, and a humidity between 40 and 60%.

Generation of transgenic parasites and genotype analyses

To generate lines for GFP-tagged SUN1 and ALLAN, a region of each gene downstream of the ATG start codon was amplified, ligated to the p277 vector, and transfected as previously described (Guttery et al., 2014). The p277 vector includes a human DHFR cassette, providing resistance to pyrimethamine. Schematic representations of the endogenous gene loci, the vector constructs, and the recombined gene loci can be found in Supplementary Figs. 1A and 6A. For parasites expressing C-terminal GFP-tagged proteins, diagnostic PCR was performed with primer 1 (Int primer) and primer 2 (ol492) to confirm integration of the GFP - targeting constructs (Supplementary Figs. 1B and 6B). The primers used to generate the mutant parasite lines can be found in Supplementary Data 3.

Gene-deletion targeting vectors for SUN1, and ALLAN were created using the pBS-DHFR plasmid. This plasmid contains polylinker sites flanking a *Toxoplasma gondii* dhfr/ts expression cassette, which provides resistance to pyrimethamine, as described previously (Saini et al., 2017). PCR primers N1511 and N1512 were used to amplify a 1,1094 bp fragment of the 5' sequence upstream of *sun1* from genomic DNA, which was then inserted into the ApaI and HindIII restriction sites upstream of the dhfr/ts cassette in the pBS-DHFR plasmid. A 776 bp fragment from the 3' flanking region of *sun1* was generated with primers N1513 and N1514, and inserted downstream of the dhfr/ts cassette using the EcoRI and XbaI restriction sites. The same approach was used for *allan*, amplifying upstream (1044 bp) and downstream (1034 bp) sequences and insertion into the pBS-DHFR plasmid. The linear targeting sequence was released from the plasmid using *Apa*I/*Xba*I digestion. A schematic representation of the endogenous *sun1* and *allan* loci, the construct and the recombined *sun1* and *allan* loci are presented in Supplementary Fig. 1A and 6A, respectively. The primers used to generate these mutant parasite lines can be found in Supplementary Table S4. A diagnostic PCR used primer 1 (IntN151_5) and primer 2 (ol248) to confirm integration of the targeting construct, and primer 3 (Int303) and primer 4 (N1514) were used to confirm deletion of the *sun1* gene (Supplementary Fig. S1B and Supplementary Table S4). Similarly, a diagnostic PCR using primer 1 (IntN153_5) and primer 2 (ol248) was used to confirm integration of the targeting construct, and primer 3 (Int307) and primer 4 (N1534) were used to confirm deletion of the *allan* gene (Supplementary Fig. S6B and Supplementary Table S4). *P. berghei* ANKA line 2.34 (for GFP-tagging), and ANKA line 507cl1 expressing GFP (for the gene-deletion) were transfected by electroporation (Janse *et al.*, 2006).

Purification of schizonts and gametocytes

Blood cells obtained from infected mice (day 4 post infection) were cultured for 11 h and 24 h at 37°C (with rotation at 100 rpm) and schizonts were purified the following day on a 60% v/v NycoDenz (in phosphate-buffered saline (PBS)) gradient, (NycoDenz stock solution: 27.6% w/v NycoDenz in 5 mM Tris-HCl (pH 7.20), 3 mM KCl, 0.3 mM EDTA).

Gametocytes were purified using a modified version of the protocol of Beetsma et al. (1998). In brief, parasites were injected into mice pre-treated with phenylhydrazine and enriched by sulfadiazine treatment two days post-infection. Blood was collected on the fourth day post-infection, and gametocyte-infected cells were purified using a 48% NycoDenz gradient (prepared with NycoDenz stock solution as above). Gametocytes were harvested from the gradient interface and washed thoroughly.

Live-cell imaging

To investigate SUN1-GFP, and ALLAN-GFP expression during erythrocytic stages, parasites cultured in schizont medium were imaged at various stages of schizogony. Purified gametocytes were assessed for GFP expression and location at different time points (0 and 1 to 15 min) post-activation in ookinete medium (RPMI 1640 medium containing 100 µM xanthurenic acid, 1% w/v sodium bicarbonate, and 20 % v/v heat inactivated foetal bovine serum [FBS]). Zygote and ookinete stages were labelled using the cy3-conjugated 13.1 antibody (red), which targets the P28 surface protein, and examined over a 24-hour period. Oocysts and sporozoites were imaged in infected mosquito guts. All images were captured with a 63× oil immersion objective on a Zeiss AxioImager M2 microscope equipped with an AxioCam ICc1 digital camera.

Western blot analysis

Purified gametocytes were placed in lysis buffer (10 mM Tris-HCl [pH 7.5], 150 mM NaCl, 0.5 mM EDTA, and 1% NP-40). The lysed samples were placed for 10 min at 95°C after adding Laemmli buffer, and then centrifuged at 13,000 g for 5 min. Samples were electrophoresed on a 4% to 12% SDS-polyacrylamide gel, and resolved proteins were transferred to nitrocellulose membrane (Amersham Biosciences). Immunoblotting was performed using the Western Breeze Chemiluminescence Anti-Rabbit kit (Invitrogen) and an anti-GFP polyclonal antibody (Invitrogen) at a dilution of 1:1,250, according to the manufacturer's instructions.

Generation of dual tagged parasite lines

The GFP (green)-tagged SUN1 or ALLAN parasites were mixed in equal numbers with mCherry (red)-tagged lines of kinetochore marker (NDC80), basal body/axoneme marker (kinesin-8B), and spindle markers (EB1 and ARK2) and injected into mice. Mosquitoes fed on these mice 4 to 5 days post-infection, when gametocytaemia was high, were monitored for oocyst development and sporozoite formation at days 14 and 21 after feeding. Infected mosquitoes were then allowed to feed on naïve mice, and after 4 to 5 days, these mice were examined for blood-stage parasitaemia by microscopy of Giemsa-stained blood smears. Some parasites expressed both GFP- and mCherry-tagged proteins in the resultant gametocytes; these cells were purified, and fluorescence microscopy images were collected as described above.

Parasite phenotype analyses

Blood samples containing approximately 50,000 SUN1-knockout, or ALLAN-knockout parasites were injected intraperitoneally (i.p.) into mice. Asexual stage parasite development and gametocyte production were monitored by microscopy on Giemsa-stained thin smears. Four to five days post-infection, gametocytes were harvested and exflagellation and ookinete conversion were examined as described above, using a Zeiss AxioImager M2 microscope. To analyze mosquito infection and parasite transmission, 30 to 50 *Anopheles stephensi* SD 500 mosquitoes were allowed to feed for 20 min on anesthetized, infected mice that had at least 15% asexual parasitaemia and a comparable gametocyte level. To assess midgut infection, approximately 15 guts were dissected from mosquitoes on days 7 and 14 post-feeding, and oocysts were counted by microscopy using a 63× oil immersion objective. On day 21 post-feeding, another 20 mosquitoes were dissected, and their guts and salivary glands were crushed separately in a loosely fitting homogenizer to release sporozoites, which were then quantified using a haemocytometer or used for imaging.

Mosquito bite-back experiments with naïve mice were conducted 21 days post-feeding, and blood smears were examined after 3 to 4 days.

Immunoprecipitation and Mass Spectrometry

Purified male gametocyte pellets from SUN1-GFP, and ALLAN-GFP parasites at 6-min post-activation. were cross-linked by 10 min incubation with 1% formaldehyde, followed by 5 min incubation in 0.125 M glycine solution and three washes with phosphate-buffered saline [PBS; pH 7.5]. WT-GFP gametocytes were used as controls. Cell lysates were prepared and immunoprecipitation was conducted using a GFP-Trap_A Kit (Chromotek) according to the manufacturer's instructions. Briefly, lysates were incubated for 2 hours with GFP-Trap_A beads at 4°C with continuous rotation, then unbound proteins were washed away, and bound proteins were digested with trypsin. The tryptic peptides were analyzed by liquid chromatography-tandem mass spectrometry. Mascot (*http://www.matrixscience.com/*) and MaxQuant (*https://www.maxquant.org/*) search engines were used for mass spectrometry data analysis. Experiments were performed in duplicate. Peptides and proteins with a minimum threshold of 95% were used for further proteomic analysis. The PlasmoDB database was used for protein annotation, and manual curation was used to classify proteins into categories relevant for SUN1/ALLAN interactions:

chromatin, nuclear pore, ER membrane and axonemal proteins. To capture co-variation of bound proteins between different experiments (comparing GFP-only with SUN1-GFP and ALLAN-GFP) we performed a principal component analysis (PCA) using unique peptide values per protein present in two replicates per experiment. Values for undetected proteins were set to 0. Values were ln (xLJ+LJ1) transformed and PCA was performed using the ClustVis webserver (settings Nipals PCA, no scaling) (Metsalu and Vilo, 2015).

AlphaFold3 modelling

3D protein structures for SUN1 and ALLAN were modelled using the AlphaFold3 webserver (*https://alphafoldserver.com/*) with standard settings (seed set to 100) (Abramson et al., 2024). To assess the stoichiometry of the complex we modelled from monomers up to decamers for both ALLAN and SUN1, using the N-terminus, the middle domain and the C-terminus of SUN1. ALLAN consistently formed higher order structures containing trimeric complexes. Although SUN1 could form higher order structures beyond trimers, we opted to model it as a trimer given the preference of ALLAN to form trimers.

Fixed Immunofluorescence Assay

SUN1-KO and ALLAN-KO gametocytes were purified, activated in ookinete medium, fixed at various time points with 4% paraformaldehyde (PFA, Sigma) diluted in microtubule (MT)-stabilizing buffer (MTSB) for 10 to 15 min, and added to poly-L-lysine coated slides. Immunocytochemistry used mouse anti-α tubulin mAb (Sigma-T9026; used at 1:1000) as primary antibody, and secondary antibody was Alexa 568 conjugated anti-mouse IgG (Invitrogen-A11004) (used at 1:1000). Slides were mounted in Vectashield with DAPI (Vector Labs) for fluorescence microscopy with a Zeiss AxioImager M2 microscope fitted with an AxioCam ICc1 digital camera.

Liver stage cultures

Human liver hepatocellular carcinoma (HepG2) cells (European Cell Culture Collection) were grown in DMEM medium, supplemented with 10% heat inactivated FBS, 0.2% $NaHCO_3$, 1% sodium pyruvate, 1% penicillin/streptomycin and 1% L-glutamine in a humidified incubator at 37LJ°C with 5% CO_2. For infection, 1LJ×LJ10^5 HepG2 cells were seeded in a 48-well culture plate. The day after seeding, sporozoites were purified following mechanical disruption from salivary glands removed from female *A. stephensi* mosquitoes infected with PbALLAN-GFP parasites, and added in infection medium to the culture maintained for 72 h.

Ultrastructure Expansion Microscopy (UExM)

Purified gametocytes were activated for different time periods, and then activation was stopped by adding 4% paraformaldehyde. Fixed cells were then attached to a 12LJmm diameter poly-D-lysine (A3890401, Gibco) coated coverslip for 10 min. Coverslips were incubated overnight in 1.4% formaldehyde (FA)/2% acrylamide (AA) at 4°C. Gelation was performed in ammonium persulfate (APS)/TEMED (10% each)/monomer solution (23% sodium acrylate; 10% AA; 0.1% BIS-AA in PBS) for 1 h at 37°C. Gels were denatured for 15 min at 37°C and 45 min at 95°C and then incubated in distilled water overnight for complete expansion. The following day, gels were washed twice in PBS for 15 min to remove excess water, and then incubated with primary antibodies at 37°C for 3 h. After washing 3 times for 10 min in PBS/ 0.1% Tween, incubation with secondary antibodies was performed for 3 h at 37°C followed by 3 washes of 10 min each in PBS/ 0.1% Tween (all antibody incubation steps were performed at 37°C with 120 to 160 rpm shaking). Directly after antibody staining, gels were incubated in 1 ml of 594 NHS-ester (Merck: 08741) diluted to 10 μg/ml in PBS for 90 min at room temperature on a shaker. The gels were then washed 3 times for 15 min with PBS/0.1% Tween and expanded overnight in ultrapure water. One cm × 1 cm pieces were cut from the expanded gel and attached to 24 mm diameter Poly-D-Lysine (A3890401, Gibco) coated coverslips to prevent the gel from sliding and avoid drifting while imaging. The primary antibody was either against α-tubulin (1:1000 dilution, Sigma-T9026), or an anti-GFP antibody (1:250, Thermo Fisher). Secondary antibodies were anti-rabbit Alexa 488, or anti-mouse Alexa 568 (Invitrogen), used at dilutions of 1:1000. Atto 594 NHS-ester (Merck 08741) was used for bulk proteome labelling. Images were acquired on Zeiss Elyra PS.1-LSM780 and CD7-LSM900, and Airyscan confocal microscopes, where 0.4 Airy unit (AU) on confocal and 0.2 AU were used along with slow scan modes. Image analysis was performed using Zeiss Zen 2012 Black edition and Fiji-Image J.

Structured Illumination Microscopy

A small volume (3 µl) of gametocyte suspension was placed on a microscope slide and covered with a long (50 × 34 mm) coverslip to obtain a very thin monolayer and immobilize the cells. Cells were scanned with an inverted microscope using a Zeiss Plan-Apochromat 63x/1.4 oil immersion or Zeiss C-Apochromat 63x/1.2 W Korr M27 water immersion objective on a Zeiss Elyra PS.1 microscope, utilizing the structured illumination microscopy (SIM) technique. The correction collar of the objective was set to 0.17 for optimal contrast. The following settings were used in SIM mode: lasers, 405 nm: 20%, 488 nm: 16%; exposure times 200 ms (Hoechst), 100 ms (GFP), three grid rotations, five phases. The bandpass filters BP 420-480 + LP 750, and BP 495-550 + LP 750 were used for the blue and green channels, respectively. Where multiple focal planes (Z-stacks) were recorded processing and channel alignment was done as described previously (Zeeshan et al., 2024).

Electron microscopy

Gametocytes activated for 8 min and 15 min were fixed in 4% glutaraldehyde in 0.1 M phosphate buffer and processed for electron microscopy (Ferguson et al., 2005). Briefly, samples were post fixed in osmium tetroxide, treated en bloc with uranyl acetate, dehydrated, and embedded in Spurr's epoxy resin. Thin sections were stained with uranyl acetate and lead citrate prior to examination in a JEOL JEM-1400 electron microscope (JEOL, United Kingdom). The experiments were done at least 3 times to capture every stage of the parasite, and 50 to 55 cells were examined for phenotypic analysis. For more details, please see the figure legends.

RNA Isolation and Quantitative Real-Time PCR (qRT-PCR) Analyses

RNA was extracted from purified gametocytes using an RNA purification kit (Stratagene). Complementary DNA (cDNA) was synthesized using an RNA-to-cDNA kit (Applied Biosystems). Gene expression was quantified from 80 ng of total RNA using the SYBR Green Fast Master Mix kit (Applied Biosystems). All primers were designed using Primer3 (Primer-BLAST, NCBI). The analysis was conducted on an Applied Biosystems 7500 Fast machine with the following cycling conditions: 95°C for 20 sec, followed by 40 cycles of 95°C for 3 sec and 60°C for 30 sec. Three technical replicates and three biological replicates were performed for each gene tested. The genes hsp70 (PBANKA_081890) and arginyl-tRNA synthetase (PBANKA_143420) were used as endogenous control reference genes. The primers used for qPCR are listed in Supplementary Table S4.

RNA-seq Analysis

Libraries were prepared from lyophilized total RNA, starting with the isolation of mRNA using the NEBNext Poly(A) mRNA Magnetic Isolation Module (NEB), followed by the NEBNext Ultra Directional RNA Library Prep Kit (NEB) as per the manufacturer's instructions. Libraries were amplified through 12 PCR cycles (12 cycles of [15 s at 98°C, 30 s at 55°C, 30 s at 62°C]) using the KAPA HiFi HotStart Ready Mix (KAPA Biosystems). Sequencing was performed on a NovaSeq 6000 DNA sequencer (Illumina), generating paired-end 100-bp reads.

FastQC [*https://www.bioinformatics.babraham.ac.uk/projects/fastqc/* was used to analyze raw read quality and the adapter sequences were removed using Trimmomatic (v0.39) [*http://www.usadellab.org/cms/?page=trimmomatic*]. The resulting reads were mapped against the P. *berghei* genome (PlasmoDB, v68) using HISAT2 (v2-2.2.1) with the --very-sensitive parameter. Uniquely mapped, properly paired reads with mapping quality of 40 or higher were retained using SAMtools (v1.19) [*http://samtools.sourceforge.net/*]. Raw read counts were determined for each gene in the *P. berghei* genome using BedTools [*https://bedtools.readthedocs.io/en/latest/* #] to intersect the aligned reads with the genome annotation. Differential expression analysis was performed using DESeq2 to call up-and down-regulated genes (FDR < 0.05 and log2 FC > 1.0). Volcano plots were made using the R package Enhanced Volcano.

Statistical Analysis

All statistical analyses were conducted using GraphPad Prism 9 (GraphPad Software). Student's t-test and/or a two-way ANOVA test were employed to assess differences between control and experimental groups. Statistical significance is indicated as *P < 0.05, **P < 0.01, ***P < 0.001, or ns for not significant. 'n' denotes the sample size in each group or the number of biological replicates. For qRT-PCR data, a multiple comparisons t-test, with post hoc Holm–Sidak test, was utilized to evaluate significant differences between wild-type and mutant parasites.

Data availability

The RNA-seq data generated in this study have been deposited in the NCBI Sequence Read Archive. Mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE partner repository. Source data are provided with this paper.

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Additional information

Contributions

R.T. conceived and coordinated the project; M.Z., I.B. and R.T. performed and analysed the live-cell imaging, knockout, and proteomics data; M.Z., R.T., I.B. and D.B. performed mice and mosquito-related work; R.Y. and D.J.P.F performed and analysed electron microscopy imaging. M.Z., S.L.P. and R.Y. performed expansion microscopy; Z.C., B.M. and S.B. performed RNA-seq analysis; Y.Y.B. and C.Y.B. did lipid analysis; A.M. performed liver stage microscopy; M.H., D.J.P.F. and S.V. performed SBF-SEM and 3D nuclear modelling. R.M. performed structured illumination microscopy; I.B. and D.B. performed qRT PCRs; A.B. performed mass spectrometry analysis; E.C.T. performed the interactomics, structural and phylogenetic analyses; S. B. performed, structural analyses; R.T., and A.A.H. supervised the project; M.Z., D.B. and R.T. managed the project. M.Z. E.C.T. and R.T. prepared the first manuscript draft. A.A.H. helped significantly in editing and writing and all authors contributed to manuscript editing and revisions.

Additional files

Fig S1

Fig S2

Fig S3

Fig S4

Fig S5

Fig S6

Fig S7

Fig S8

Fig S9

Fig S10

Fig S11

SV1

SV2

Table S1

Table S2

Table S3

Table S4

References

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Author information

Mohammad Zeeshan[+]
University of Nottingham, School of Life Sciences, Nottingham, United Kingdom
ORCID iD: 0000-0002-6185-403X

[+]These authors contributed equally

Igor Blatov[+]
University of Nottingham, School of Life Sciences, Nottingham, United Kingdom

[+]These authors contributed equally

Ryuji Yanase[#]
University of Nottingham, School of Life Sciences, Nottingham, United Kingdom
ORCID iD: 0000-0001-9419-398X

[#]These authors contributed equally

David JP Ferguson[#]
Oxford Brookes University, Department of Biological and Medical Sciences, Oxford, United Kingdom

[#]These authors contributed equally

Sarah L Pashley
University of Nottingham, School of Life Sciences, Nottingham, United Kingdom

Zeinab Chahine
Department of Molecular, Cell and Systems Biology, University of California Riverside, Riverside, United States

Yoshiki Yamaryo Botté
Apicolipid Team, Institute for Advanced Biosciences, CNRS UMR5309, Université Grenoble Alpes, INSERM U1209, Grenoble, France

Akancha Mishra
University of Nottingham, School of Life Sciences, Nottingham, United Kingdom

Baptiste Marché
Department of Molecular, Cell and Systems Biology, University of California Riverside, Riverside, United States

Suhani Bhanvadia
Department of Molecular, Cell and Systems Biology, University of California Riverside, Riverside, United States

Molly Hair
Oxford Brookes University, Department of Biological and Medical Sciences, Oxford, United Kingdom

Sagar Batra
University of Nottingham, School of Life Sciences, Nottingham, United Kingdom

Robert Markus
University of Nottingham, School of Life Sciences, Nottingham, United Kingdom

Declan Brady
University of Nottingham, School of Life Sciences, Nottingham, United Kingdom

Andrew Bottrill
School of Life Sciences, Gibbet Hill Campus, University of Warwick, Coventry, United Kingdom

Sue Vaughan
Oxford Brookes University, Department of Biological and Medical Sciences, Oxford, United Kingdom

Cyrille Y Botté
Apicolipid Team, Institute for Advanced Biosciences, CNRS UMR5309, Université Grenoble Alpes, INSERM U1209, Grenoble, France

Karine Le Roch
Department of Molecular, Cell and Systems Biology, University of California Riverside, Riverside, United States

Anthony A Holder
Malaria Parasitology Laboratory, The Francis Crick Institute, London, United Kingdom
ORCID iD: 0000-0002-8490-6058

Eelco C Tromer
Cell Biochemistry, Groningen Biomolecular Sciences and Biotechnology Institute, Faculty of Science and Engineering, University of Groningen, Groningen, Netherlands

Rita Tewari
University of Nottingham, School of Life Sciences, Nottingham, United Kingdom
ORCID iD: 0000-0003-3943-1847

For correspondence: Rita.Tewari@nottingham.ac.uk

Editors

Reviewing Editor
Dominique Soldati-Favre
University of Geneva, Geneva, Switzerland

Senior Editor
Dominique Soldati-Favre
University of Geneva, Geneva, Switzerland

Reviewer #1 (Public review):

Summary:

Activated male Plasmodium gametocytes undergo very rapid nuclear division, while keeping the nuclear envelope intact. There is interest in how events inside the nucleus are co-ordinated with events in the parasite cytoplasm, to ensure that each nucleus is packaged into a nascent male gamete.

This manuscript by Zeeshan et al describes the organisation of a nuclear membrane bridging protein, SUN1, during nuclear division. SUN1 is expected from studies in other organisms to be a component of a bridging complex (LINC) that connects the inner nuclear membrane to the outer nuclear membrane, and from there to the cytoplasmic microtubule-organising centres, the centrosome and the basal body.

The authors show that knockout of the SUN1 in gametocytes leads to severe disruption of the mitotic spindle and failure of the basal bodies to segregate. The authors show convincingly that functional SUN1 is required for male gamete formation and subsequent oocyst development.

The authors identified several SUN1-interacting proteins, thus providing information about the nuclear membrane bridging machinery.

Strengths:

The authors have used state of the art imaging, genetic manipulation and immunoprecipitation approaches.

Weaknesses:

Technical limitations of some of the methods used make it difficult to interpret some of the micrographs.

From studies in other organisms, a protein called KASH is a critical component the bridging complex (LINC). That is, KASH links SUN1 to the outer nuclear membrane. The authors undertook a gene sequence analysis that reveals that Plasmodium lacks a KASH homologue. Thus, further work is needed to identify the functional equivalent of KASH, to understand bridging machinery in Plasmodium.

Comments on revised version:

The authors have addressed the comments and suggestions that I provided as part of a Review Commons assessment.

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Reviewer #2 (Public review):

Zeeshan et al. investigate the function of the protein SUN1, a proposed nuclear envelope protein linking nuclear and cytoplasmic cytoskeleton, during the rapid male gametogenesis of the rodent malaria parasite Plasmodium berghei. They reveal that SUN1 localises to the nuclear envelope (NE) in male and female gametes and show that the male NE has unexpectedly high dynamics during the rapid process of gametogenesis. Using expansion microscopy, the authors find that SUN1 is enriched at the neck of the bipartite MTOC that links the intranuclear spindle to the basal bodies of the cytoplasmic axonemes. Upon deletion of SUN1, the basal bodies of the eight axonemes fail to segregate, no spindle is formed, and emerging gametes are anucleated, leading to a complete block in transmission. By interactomics the authors identify a divergent allantoicase-like protein, ALLAN, as a main interaction partner of SUN1 and further show that ALLAN deletion largely phenocopies the effect of SUN1.

Overall, the authors use an extensive array of fluorescence and electron microscopy techniques as well as interactomics to convincingly demonstrate that SUN1 and ALLAN play a role in maintaining the structural integrity of the bipartite MTOC during the rapid rounds of endomitosis in male gametogenesis.

Two suggestions for improvement of the work remain:

(1) Lipidomic analysis of WT and SUN1-knockout gametocytes before and after activation resulted in only minor changes in some lipid species. Without statistical analysis, it remains unclear if these changes are statistically significant and not rather due to expected biological variability. While the authors clearly toned down their conclusions in the revised manuscript, some phrasings in the results and the discussion still suggest that gametocyte activation and/or SUN1-knockout affects lipid composition. Similarly, some phrases suggest that SUN1 is responsible for the observed loops and folds in the NE and that SUN1 KO affects the NE dynamics. Currently, I do not think that the data supports these statements.

(2) It is interesting to note that ALLAN has a much more specific localisation to basal bodies than SUN1, which is located to the entire nuclear envelope. Knock out of ALLAN also exhibits a milder (but still striking) phenotype than knockout of SUN1. These observations suggest that SUN1 has additional roles in male gametogenesis besides its interaction with ALLAN, which could be discussed a bit more.

This study uses extensive microscopy and genetics to characterise an unusual SUN1-ALLAN complex, thus providing new insights into the molecular events during Plasmodium male gametogenesis, especially how the intranuclear events (spindle formation and mitosis) are linked to the cytoplasmic separation of the axonemes. The characterisation of the mutants reveals an interesting phenotype, showing that SUN1 and ALLAN are localised to and maintain the neck region of the bipartite MTOC. The authors here confirm and expand the previous knowledge about SUN1 in P. berghei, adding more detail to its localisation and dynamics, and further characterise the interaction partner ALLAN. Given the evolutionary divergence of Plasmodium, these results are interesting not only for parasitologists, but also for more general cell biologists.

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Author response:

> ***Reviewer #1 (Evidence, reproducibility and clarity):***
>
> *Minor comments:*
>
> *In the results section (lines 498-499), the authors describe free kinetochores in many cells without associated spindle microtubules. However, some nuclei appear to have kinetochores, as presented in Figure 6. Could the authors clarify how this conclusion was derived using transmission electron microscopy (TEM) without serial sectioning, as this is not explicitly mentioned in the materials and methods?*

We observed free kinetochores in the ALLAN-KO parasites with no associated spindle microtubules (see Fig. 6Gh), while kinetochores are attached to spindle microtubules in WT-GFP cells (see Fig. 6Gc). To provide further evidence we analysed additional images and found that ALLAN-KO cells have free kinetochores in the centre of nucleus, unattached to spindle microtubules. We provide some more images clearly showing free kinetochores in these cells (new supplementary Fig. S11).

However, in the ALLAN mutant, this difference is not absolute: in a search of over 50 cells, one example of a cell with a “normal” nuclear spindle and attached kinetochores was observed.

The use of serial sectioning has limitations for examining small structures like kinetochores in whole cells. The limitations of the various techniques (for example, SBF-SEM vs tomography) are highlighted in our previous study (Hair et al 2022; PMID: 38092766), and we consider that examining a population of randomly sectioned cells provides a better understanding of the overall incidence of specific features.

> *Discussion Section:*
>
> *Could the authors expand on why SUN1 and ALLAN are not required during asexual replication, even though they play essential roles during male gametogenesis?*

We observed no phenotype in asexual blood stage parasites associated with the *sun1* and *allan* gene deletions. Several other *Plasmodium berghei* gene knockout parasites with a phenotype in sexual stages, for example CDPK4 (PMID: 15137943), SRPK (PMID: 20951971), PPKL (PMID: 23028336) and kinesin-5 (PMID: 33154955) have no phenotype in blood stages, so perhaps this is not surprising. One explanation may be the substantial differences in the mode of cell division between these two stages. Asexual blood stages produce new progeny (merozoites) over 24 hours with closed mitosis and asynchronous karyokinesis during schizogony, while male gametogenesis is a rapid process, completed within 15 min to produce eight flagellated gametes. During male gametogenesis the nuclear envelope must expand to accommodate the increased DNA content (from 1N to 8N) before cytokinesis. Furthermore, male gametogenesis is the only stage of the life cycle to make flagella, and axonemes must be assembled in the cytoplasm to produce the flagellated motile male gametes at the end of the process. Thus, these two stages of parasite development have some very different and specific features.

> *Lines 611-613 states: "These loops serve as structural hubs for spindle assembly and kinetochore attachment at the nuclear MTOC, separating nuclear and cytoplasmic compartments." Could the authors elaborate on the evidence supporting this statement?*

We observed the loops/folds in the nuclear envelope (NE) as revealed by SUN1-GFP and 3D TEM images during male gametogenesis. These folds/loops occur mainly in the vicinity of the nuclear MTOC where the spindles are assembled (as visualised by EB1 fluorescence) and attached to kinetochores (as visualised by NDC80 fluorescence). These loops/folds may form due to the contraction of the spindle pole back to the nuclear periphery, inducing distortion of the NE. Since there is no physical segregation of chromosomes during the three rounds of mitosis (DNA increasing from 1N to 8N), we suggest that these folds provide additional space for spindle and kinetochore dynamics within an intact NE to maintain separation from the cytoplasm (as shown by location of kinesin-8B).

> *In lines 621-622, the authors suggest that ALLAN may have a broader role in NE remodelling across the parasite's lifecycle. Could they reflect on or remind readers of the finding that ALLAN is not essential during the asexual stage?*

ALLAN-GFP is expressed throughout the parasite life cycle but as the reviewer points out, a functional role is more pronounced during male gametogenesis. This does not mean that it has no role at other stages of the life cycle even if there is no obvious phenotype following deletion of the gene during the asexual blood stage. The fact that ALLAN is not essential during the asexual blood stage is noted in lines 628-29.

> ***Reviewer #2 (Evidence, reproducibility and clarity):***
>
> *Introduction*
>
> *Line 63: The authors stat: "NE is integral to mitosis, supporting spindle formation, kinetochore attachment, and chromosome segregation..". Seemingly at odds, they also say (Line 69) that 'open' "mitosis is "characterized by complete NE disassembly".*

> *The authors could explain better the ideas presented in their quoted review from Dey and Baum, which points out that truly 'open' and 'closed' topologies may not exist and that even in 'open' mitosis, remnants of the NE may help support the mitotic spindle.*

We have modified the sentence in which we discuss current opinions about ‘open’ and ‘closed’ mitosis. It is believed that there is no complete disassembly of the NE during open mitosis and no completely intact NE during closed mitosis, respectively. In fact, the NE plays a critical role in the different modes of mitosis during MTOC organisation and spindle dynamics. Please see the modified lines 64-71.

> *Results*
>
> *Fig 7 is the final figure; but would be more useful upfront.*

We have provided a new introductory figure (Fig 1) showing a schematic of conventional /canonical LINC complexes and evidence of SUN protein functions in model eukaryotes and compare them to what is known in apicomplexans.

> *Fig 1D. The authors generated a C-terminal GFP-tagged SUN1 transfectants and used ultrastructure expansion microscopy (U-ExM) and structured illumination microscopy (SIM) to examine SUN1-GFP in male gametocytes post-activation. The immuno-labelling of SUN1-GFP in these fixed cells appears very different to the live cell images of SUN1-GFP. The labelling profile comprises distinct punctate structures (particularly in the U-ExM images), suggesting that paraformaldehyde fixation process, followed by the addition of the primary and secondary antibodies has caused coalescing of the SUN1-GFP signal into particular regions within the NE.*

We agree with the reviewer. Fixation with paraformaldehyde (PFA) results in a coalescence of the SUN1-GFP signal. We have also tried methanol fixation (see new Fig. S2), but a similar problem was encountered.

> *Given these fixation issues, the suggestion that the SUN1-GFP signal is concentrated at the BB/ nuclear MTOC and "enriched near spindle poles" needs further support.*
>
> *These statements seem at odd with the data for live cell imaging where the SUN1-GFP seems evenly distributed around the nuclear periphery. Can the observation be quantitated by calculating the percentage of BB/ nuclear MTOC structures with associated SUN1-GFP puncta? If not, I am not convinced these data help understand the molecular events.*

We agree with the reviewer that whilst the live cell imaging showed an even distribution of SUN1-GFP signal, after fixation with either PFA or methanol, then SUN1-GFP puncta are observed in addition to the peripheral location around the stained DNA (Hoechst) (See Fig. S2; puncta are indicated by arrows). These SUN1-GFP labelled puncta were observed at the junction of the nuclear MTOC and the basal body (Fig. 2F). Quantification of the distribution showed that these SUN1-GFP puncta are associated with nuclear MTOC in more than 90 % of cells (18 cells examined). Live cell imaging of the dual labelled parasites; SUN1xkinesin-8B (Fig. 2H) and SUN1x EB1 (Fig. 2I) provides further support for the association of SUN1-GFP puncta with BB (kinesin-8B) /nuclear MTOC (EB1).

> *The authors then generated dual transfectants and examined the relative locations of different markers in live cells. These data are more informative.*

> *The authors state; " ..SUN1-GFP marked the NE with strong signals located near the nuclear MTOCs situated between the BB tetrads". The nuclear MTOCs are not labelled in this experiment. The SUN1-GFP signal between the kinesin-8B puncta is evident as small puncta on regions of NE distortion. I would prefer to not describe this signal as "strong". The signal is stronger in other regions of the NE.*

We have modified the sentence on line 213 to accommodate this suggestion.

> *Line 219. The authors state; "..SUN1-GFP is partially colocalized with spindle poles as indicated by EB1,.. it shows no overlap with kinetochores (NDC80)." The authors should provide an analysis of the level of overlap at a pixel by pixel level to support this statement.*

We now provide the overlap at a pixel-by-pixel level for representative images, and we have quantified more cells (n>30), as documented in the new Fig. S4A. We have also modified the sentence on line 219 to reflect these additions.

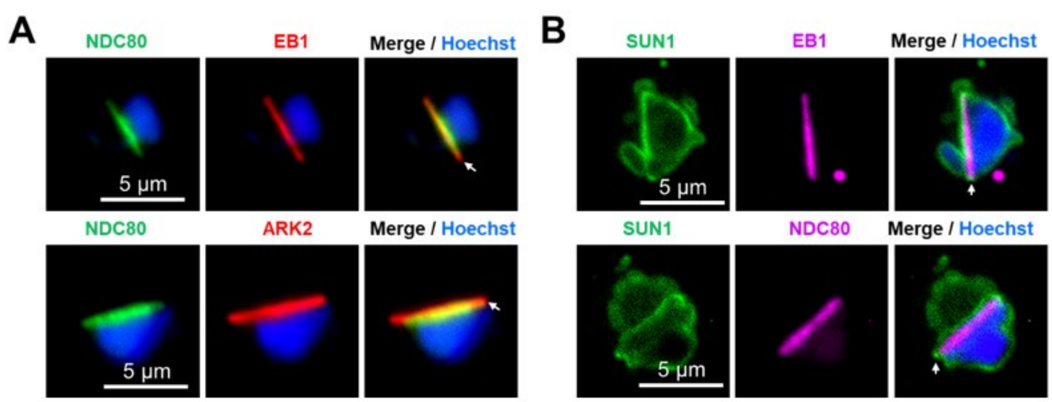
> *The SUN1 construct is C-terminally GFP-tagged. By analogy with human SUN1, the C-terminal SUN domain is expected to be in the NE lumen. That is in a different compartment to EB1, which is located in the nuclear lumen (on the spindle). Thus, the overlap of signal is expected to be minimal.*

We agree with the reviewer that the overlap between EB1 and Sun1 signals is expected to be minimal. We have quantified the data and included it in Supplementary Fig. S4A.

> *Similarly, given that EB1 and NDC80 are known to occupy overlapping locations on the spindle, it seems unlikely that SUN1 can overlap with one and not the other.*

We agree with the reviewer's analysis that EB1 and NDC80 occupy overlapping locations on the spindle, although the length of NDC80 is less at the ends of spindles (see Author response image 1A) as shown in our previous study where we compared the locations of two spindle proteins, ARK2 and EB1, with that of NDC80 (Zeeshan et al, 2022; PMID: 37704606). In the present study we observed that Sun1-GFP partially overlaps with EB1 at the ends of the spindle, but not with NDC80. Please see Author response image 1B.

Author response image 1.



> *I note on Line 609, the authors state "Our study demonstrates that SUN1 is primarily localized to the nuclear side of the NE.." As per Fig 7D, and as discussed above, the bulk*

> *of the protein, including the SUN1 domain, is located in the space between the INM and the ONM.*

We appreciate the reviewer's correction; we have now modified the sentence to indicate that the protein is largely localized in the space between the INM and the ONM on line 617.

> *Interestingly, as the authors point out, nuclear membrane loops are evident around EB1 and NDC80 focal regions. The data suggests that the contraction of the spindle pole back to the nuclear periphery induces distortion of the NE.*

We agree with the reviewer's suggestion that the data indicate that contraction of spindle poles back to the nuclear periphery may induce distortion of the NE.

> *The author should discuss further the overlap of findings of this study with that from a recent manuscript (https://doi.org/10.1016/j.cels.2024.10.008). That Sayers et al. study identified a complex of SUN1 and ALLC1 as essential for male fertility in P. berghei. Sayers et al. also provide evidence that this complex particulate in the linkage of the MTOC to the NE and is needed for correct mitotic spindle formation during male gametogenesis.*

We thank the reviewer for this suggestion. The study by Sayers et al, (2024) was published while our manuscript was under preparation. It was interesting to see that these complementary studies have similar findings about the role of SUN1 and the novel complex of SUN1-ALLAN. Our study contains a more detailed, in-depth analysis both by Expansion and TEM of SUN1. We include additional studies on the role of ALLAN. We discuss the overlap in the findings of the two studies in lines 590-605.

> *While the work is interesting, the conclusions may need to be tempered. The authors suggestion that in the absence of KASH-domain proteins, the SUN1-ALLAN complex forms a non-canonical LINC complex (that is, a connection across the NE), that "achieves precise nuclear and cytoskeletal coordination".*

We have toned down the wording of this conclusion in lines 665-677.

> *In other organisms, KASH interacts with the C-terminal domain on SUN1, which as mentioned above is located between the INM and ONM. By contrast, ALLAN interacts with the N-terminal domain of SUN1, which is located in the nuclear lumen. The SUN1-ALLAN interaction is clearly of interest, and ALLAN might replace some of the roles of lamins. However, the protein that functionally replaces KASH (i.e. links SUN1 to the ONM) remains unidentified.*

We agree with reviewer, and future studies will need to focus on identifying the KASH replacement that links SUN1 to the ONM.

> *It may also be premature to suggest that the SUN1-ALLAN complex is promising target for blocking malaria transmission. How would it be targeted?*

We have deleted the sentence that raised this suggestion.

> *While the above datasets are interesting and internally consistent, there are two other aspects of the manuscript that need further development before they can usefully contribute to the molecular story.*

> *The authors undertook a transcriptomic analysis of Δsun1 and WT gametocytes, at 8 and 30 min post-activation, revealing moderate changes (~2-fold change) in different genes. GO-based analysis suggested up-regulation of genes involved in lipid metabolism. Given the modest changes, it may not be correct to conclude that "lipid metabolism and microtubule function may be critical functions for gametogenesis that can be perturbed by sun1 deletion." These changes may simply be a consequence of the stalled male gametocyte development.*

Following the reviewer's suggestion we have moved these data to the supplementary information (Fig. S5D-I) and toned down their discussion in the results and discussion sections.

> *The authors have then undertaken a detailed lipid analysis of the Δsun1 and WT gametocytes, before and after activation. Substantial changes in lipid metabolites might not be expected in such a short period of time. And indeed, the changes appear minimal. Similarly, there are only minor changes in a few lipid sub-classes between Δsun1 and WT gametocytes. In my opinion, the data are not sufficient to support the authors conclusion that "SUN1 plays a crucial role, linking lipid metabolism to NE remodelling and gamete formation."*

In agreement with the reviewer's comments we have moved these data to supplementary information (Fig. S6) and substantially toned down the conclusions based on these findings.

> ***Reviewer #3 (Evidence, reproducibility and clarity):***
>
> *Major comments:*
>
> *My main concern with this manuscript is that the authors do conclude not only that SUN1 is important for spindle formation and basal body segregation, but also that it influences for lipid metabolism and NE dynamics. I don't think the data supports this conclusion, for several reasons listed below. I would suggest to remove this claim from the manuscript or at least tone it down unless more supporting data are provided, in particular showing any change in NE dynamics in the SUN1-KO. Instead I would recommend to focus on the more interesting role of SUN1-ALLAN in bipartite MTOC organisation, which likely explains all observed phenotypes (including those in later stages of the parasite life cycle). In addition, some aspects of the knockout phenotype should be quantified to a bit deeper level.*
>
> *In more detail:*
>
> *- The lipidomics analysis is clearly the weakest point of the manuscript: The authors state that there are significant changes in some lipid populations between WT and sun1-KO, and between activated and non-activated cells, yet no statistical analysis is shown and the error bars are quite high compared to only minor changes in the means. For some discussed lipids, the result text does not match the graphs, e.g. PA, where the increase upon activation is more pronounced in the SUN1-KO vs WT (contrary to the text), or MAG, which is reduced in the SUN1-KO vs WT (contrary to the text). I don't see the discussed changes in arachidonic acid levels and myristic acid levels in the data either. Even if the authors find after analysis some statistically significant differences between some groups, they should carefully discuss the biological significance of these differences. As it is, I do not think the presented data warrants the conclusion that deletion of SUN1 changes lipid homeostasis, but rather shows that overall lipid homeostasis is not majorly affected by gametogenesis or SUN1 deletion. As a minor comment, if you decide to keep the lipidomics analysis in the manuscript, please state how many replicates were done.*

As detailed above we have moved the lipidomics data to supplementary information (Fig. S6) and substantially toned down the discussion of these data in the results and discussion sections.

> *- I can't quite follow the logic why the authors performed transcriptomic analysis of the SUN1 and how they chose their time points. Their data up to this point indicate that SUN1 has a structural or coordinating role in the bipartite MTOC during male gametogenesis. Based on that it is rather unlikely that SUN1 KO directly leads to transcriptional changes within the 8 min of exflagellation. Isn't it more likely that transcriptional differences are purely a downstream effect of incomplete/failed gametogenesis? This is particularly true for the comparison at 30 min, which compares a mixture of exflagellated/emerged gametes and zygotes in WT to a mixture of aberrant, arrested gametes in the knockout, which will likely not give any meaningful insight. The by far most significant GO-term is then also nuclear-transcribed mRNA catabolic process, which is likely not related at all to SUN1 function (and the authors do not even comment on this in the main text). I would therefore suggest removing the 30 min data set from this manuscript. As a minor point, I would suggest highlighting some of the top de-regulated gene IDs in the volcano plots and stating their function. Also, please state how you prepared the cells for the transcriptomes and in how many replicates this was done.*

As suggested by the reviewer we have removed the 30 min post activation data from the manuscript. We have also moved the rest of the transcriptomics data to supplementary information (Fig. S5) and toned down the presentation of this aspect of the work in the results and discussion sections.

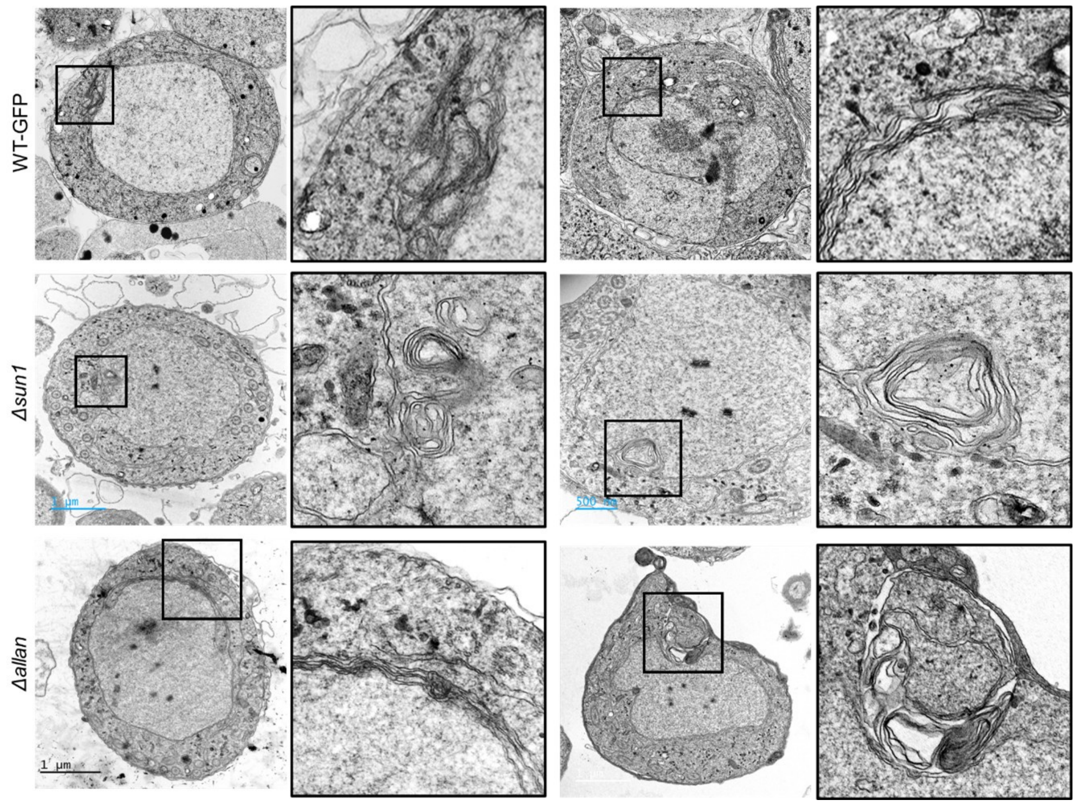
> *- Live-cell imaging of SUN1-GFP does nicely visualise the NE during gametogenesis, showing a highly dynamic NE forming loops and folds, which is very exciting to see. It would be beneficial to also show a video from the life-cell imaging.*

We have now added videos to the manuscript as suggested by the reviewer. Please see the supplementary Videos S1 and S2.

> *In their discussion, the authors state multiple times that NE dynamics are changed upon SUN1 KO. Yet, they do not provide data supporting this claim, i.e. that the extended loops and folds found in the nuclear envelope during gametogenesis are affected in any way by the knockout of SUN1 or ALLAN. What happens to the NE in absence of SUN1? Are there less loops and folds? In absence of a reliable NE marker this may not be entirely easy to address, but at least some SBF-SEM images of the sun1-KO gametocytes could provide insight.*

It was difficult to provide SBF-SEM images as that work is beyond the scope of this manuscript. We will consider this approach in our future work. We re-examined many of our TEM images of SUN1-KO and ALLAN-KO parasites and did find some micrographs showing aberrant nuclear membrane folding (<5%) (Please see Author response image 2). However, we also observed similar structures in some of the WT-GFP samples (<5%), so we do not think this is a strong phenotype of the SUN1 or ALLAN mutants.

Author response image 2.



> *- I think the exciting part of the manuscript is the cell biological role of SUN1 on male gametogenesis, which could be carved out a bit more by a more detailed phenotyping. Specifically it would be good to quantify*
>
> *(1) If DNA replication to an octoploid state still occurs in SUN1-KO and ALLAN-KO,*

DNA replication is not affected in the SUN1-KO and ALLAN-KO mutants: DNA content increases to 8N (data added in Fig. 3J and Fig. S10F).

> *(2) The proportion of anucleated gametes in WT and the KO lines*

We have added these data in Fig. 3K and Fig. S10G

> *(3) A quantification of the BB clustering phenotype (in which proportion of cells do the authors see this phenotype). This could be addressed by simple fixed immunofluorescence images of the respective WT/KO lines at various time points after activation (or possibly by reanalysis of the already obtained images) and would really improve the manuscript.*

We have reanalysed the BB clustering phenotype and added the quantitative data in Fig. 4E and Fig. S7.

> *Especially the claim that emerged SUN1-KO gametes lack a nucleus is currently only based on single slices of few TEM cells and would benefit from a more thorough quantification in both SUN1- and ALLAN-Kos*

We have examined many microgametes (100+ sections). In WT parasites a small proportion of gametes can appear to lack a nucleus if it does not extend all the way to the apical and basal ends (Hair et al. 2022). However, the proportion of microgametes that appear to lack a nucleus (no nucleus seen in any section) was much higher in the SUN1 mutant. In contrast, this difference was not as clear cut in the ALLAN mutant with a small proportion of intact (with axoneme and nucleus) microgametes being observed.

We have done additional analysis of male gametes, looking for the presence of the nucleus by live cell imaging after DNA staining with Hoechst. These data are added in Fig. 3K (for Sun1-KO) and Fig. S10G (for Allan-KO).

> *- The TEM suggests that in the SUN1-KO, kinetochores are free in the nucleus. Are all kinetochores free or do some still associate to a (minor/incorrectly formed) spindle? The authors could address this by tagging NDC80 in the KO lines.*

Our observation and quantification of the data indicated that 100% of kinetochores were attached to spindle microtubules and that 0% were unattached kinetochores in the WT parasites. However, the exact opposite was found for the SUN1 mutant with 100% unattached kinetochores and 0% attached. The result was not quite as clear cut in the ALLAN mutant, with 98% unattached and 2% attached. An important observation was the lack of separation of the nuclear poles and any spindle formation. Spindle formation was never or very rarely observed in the mutants.

> *- Finally, I think it is curious that in contrast to SUN1, ALLAN seems to be less important, with some KO parasite completing the life cycle. Maybe a more detailed phenotyping as above gives some more hints to where the phenotypic difference between the two proteins lies. I would assume some ALLAN-KO cells can still segregate the basal body. Can the authors speculate/discuss in more detail why these two proteins seems to have slightly different phenotypes?*

We agree with the reviewer. Overall, the ALLAN-KO has a less prominent phenotype than that of the Sun1-KO. The main difference is that in the ALLAN-KO mutant some basal body segregation can occur, leading to the production of some fertile microgametocytes, and ookinetes, and oocyst formation (Fig. 8). Approximately 5% of oocysts sporulated to release infective sporozoites that could infect mice in bite back experiments and complete the life cycle. In contrast the Sun1-KO mutant made no healthy oocysts, or infective sporozoites, and could not complete the life cycle in bite back experiments. We have analysed the phenotype in detail and provide quantitative data for gametocyte stages by EM and ExM in Figs. 4 and S8 (SUN1) and Figs. 7 and S11 (ALLAN). We have also performed detailed analysis of oocyst and sporozoite stages and included the data in Fig. 3 (SUN1) and S10 (ALLAN).

Based on the location, and functional and interactome data, we think that SUN1 plays a central role in coordinating nucleoplasm and cytoplasmic events as a key component of the nuclear membrane lumen, whereas ALLAN is located in the nucleoplasm. Deleting the SUN1 gene may disrupt the connection between INM and ONM whereas the deletion of ALLAN may affect only the INM.

> *Some additional points where the data is not entirely sound yet or could be improved:*
>
> *- Localisation of SUN1: There seems to be a discrepancy between SUN1-GFP location as observed by live cell microscopy, and by Expansion Microscopy (ExM), similar for ALLAN-GFP. By live-cell microscopy, the SUN1 localisation is much more evenly distributed around the NE, while the localisation in ExM is much more punctuated, and e.g. in Figure 1E seems to be within the nucleus. Do the authors have an explanation for this? Also, in*

> *Fig. 1D there are two GFP foci at the cell periphery (bottom left of the image), which I would think are not SUN1-Foci, as they seem to be outside of the cell. Is the antibody specific? Was there a negative control done for the antibody (WT cells stained with GFP antibodies after ExM)?*

High resolution SIM and expansion microscopy showed that the SUN1-GFP molecules coalesce to form puncta, in contrast to the more uniform distribution observed by live cell imaging. This apparent difference may be due to a better resolution that could not be achieved by live cell imaging. We agree with the reviewer that the two green foci are outside of the cell. As a negative control we have used WT-ANKA cells (which contain no GFP) and the anti-GFP antibody, which gave no signal. This confirms the specificity of the antibody (please see the new Fig. S3).

> *- The authors argue that SIM gave unexpected results due to PFA fixation leading to collapse of the NE loops. However, they also fix their ExM cells and their EM cells with PFA and do not observe a collapse, at least from what I see in the two presented images and in the 3D reconstruction. Is there something else different in the sample preparation?*

There was no difference in the fixation process for samples examined by SIM and ExM, but we used an anti-GFP antibody in ExM to visualise the SUN1-GFP, while in SIM the images of GFP signal were collected directly after fixation. We used both PFA and methanol as fixative, and both methods showed a coalescing of the SUN1-GFP signal (please see the new Fig. S2 and S3).

> *Can the authors trace their NE in ExM according to the NHS-Ester signal?*

We could trace the NE in the ExM by the NHS-ester signal and observed that the SUN1-GFP signal was largely coincident with the NE (Please see the new Fig. S3B).

> *- Fig 2D: It would be good to not just show images of oocysts but actually quantify their size from images. Also, have the authors determined the sporozoite numbers in SUN1-KO?*

We have measured oocyst size (data added in new Fig. 3) and added the sporozoite quantification data in Fig. 3D.

> *- Line 481-483: the authors state that oocyst size is reduced in ALLAN-KO but do not show the data. Please quantify oocyst size or at least show representative images. Also the drastic decrease in sporozoite numbers (Fig. 6D, E) is not mentioned in the text. Please add reference to Fig S7D when talking about the bite back data.*

We have added the oocyst size data in Fig. S10. We mention the changes in sporozoite numbers (now shown in Fig. 7D, E), and refer to the bite back data shown in current Fig. 7E.

> *- Fig S1C, 6C: Both WB images are stitched, but this is not clearly indicated e.g. by leaving a small gap between the lanes. Also please show a loading control along with the western blots. Also there seems to be a (unspecific?) band in the control, running at the same height as Allan-GFP WB. What exactly is the control?*

We have provided the original blot showing the bands of ALLAN-GFP and SUN1-GFP. As a positive control, we used an RNA associated protein (RAP-GFP) that is highly expressed in Plasmodium and regularly used in our lab for this purpose.

- Regarding the crossing experiment: The authors conclude from this cross that SUN1 is only needed in males, yet for this conclusion they would need to also show that a cross with a female line does not rescue the phenotype. The authors should repeat the cross with a male-deficient line to really test if the phenotype is an exclusively male phenotype. In addition, line 270-272 states that no oocysts/sporozoites were detected in sun1-ko and nek4-ko parasites. However, the figure 2E shows only oocysts, not sporozoites, and shows also that sun1-ko does form oocysts, albeit dead ones.

We have now performed the experiment of crossing the Sun1-KO parasite line with a male deficient line (Hap2-KO) and added the data in Fig. 3I. We have added images showing sporozoites in oocysts.

- In Fig S1 the authors show that they also generated a SUN1-mCherry line, yet they do not use it in any of the presented experiments (unless I missed it). Would it be beneficial to cross the SUN1-mCherry line with the Allan1-GFP line to test colocalisation (possibly also by expansion microscopy)?

We did generate a SUN1-mCherry line, with the intent to cross ALLAN-GFP and SUN1-mCherry lines and observe the co-location of the proteins. Despite multiple attempts this cross was unsuccessful. This may have been due to their close proximity such that the addition of both GFP and mCherry was difficult to facilitate a proper protein-protein interaction between either of the proteins.

- Line 498: "In a significant proportion of cells" - What was the proportion of cells, and what does significant mean in this context?

Approximately 67% of cells showed the clumping of BBs. We have now added the numbers in Figs. 6H and S11I.

- The authors should discuss a bit more how their work relates to the work of Sayers et al. 2024, which also identified the SUN1-ALLAN complex. The paper is cited, but only very briefly commented on.

We have extended this discussion now in lines 590-605.

Suggestions how to improve the writing and data presentation.

- General presentation of microscopy images: Considering that large parts of the manuscript are based on microscopy data, their presentation could be improved. Single-channel microscopy images would benefit from being depicted in gray scale instead of color, which would make it easier to see the structures and intensities (especially for blue channels).

Whilst we agree with the reviewer, sometimes it is difficult to see the features in the merged images. Therefore, we would like to request to be allowed to retain the colours, which can be easily followed in both individual and merged images.

Also, it would be good to harmonize in which panels arrows are shown (e.g. Fig 1G, where some white arrows are in the SUN1-GFP panel, while others are in the merge panel, but they presumably indicate the same thing.). At the same time, Fig 1H doesn't have any with arrows, even though the figure legend states so.

We apologise for this lack of consistency, and we have now added arrows wherever they are missing to harmonise in the presentations.

> *Fig 3A and S4 show the same experiment but are coloured in different colours (NHS-Eester in green vs grey scale).*
>
> *- Are the scale bars of all expansion microscopy images adjusted for the expansion factor?*

Yes, the scale bars are adjusted accordingly.

> *- The figure legends would benefit from streamlining, as they have very different style between figures (eg Fig. 6 which has a concise figure legend vs microscopy figures where figure legends are very long and describe not only the figure but the results)*

The figure legends have been streamlined, with removal of the description of results.

> *- Line 155-156: The text makes it sound like the expression only happens after activation. is that the case? Are these images activated or non-activated gametocytes?*

They are expressed before activation, but the signal intensifies after activation. Images from before and after activation of gametocytes have been added in Fig. S1F.

> *- Line 267: Reference to the original nek4-KO paper missing*

This reference is now included.

> *- Line 301: The reference to Figure 2J seems to be a bit arbitrarily placed. Also, this schematic of lipid metabolism is never discussed in relation to the transcriptomic or lipidomic data.*

We have moved these data to supplementary information and modified the text.

> *- Line 347-349 states that gametes emerged, but the referenced figure shows activated gametocytes before exflagellation.*

We have corrected the text to the start of exflagellation.

> *- Line 588: Spelling mistake in SUN1-domain*

Corrected.

> *- Line 726/731: i missing in anti-GFP*

Corrected.

> *- Line 787-789: statement of scale bar and number of cells imaged is not at the right position in the figure legend.*

Moved to right place

> *- Line 779, 783: "shades of green" should be just "green". Same goes for line 986, 989 with "shades of grey"*

Changed.

> *- Line 974, 976: please correct to WT-GFP and dsun1*

Corrected.

> *- Line 1041, 1044: WT-GFP instead of WTGFP.*

Corrected to WT-GFP.

> *- Fig 1B, D, E, Fig S1G, H: What are the time points of imaging?*

We have added the time points to the images in these figures.

> *- Fig 1D/Line 727: the scale of the scale bar on the inset is missing.*

We have added the scale bar.

> *- Fig 3 E-G and 6H-J: Please indicate total number of cells/images analysed per quantification, either in the graphs themselves or in the figure legend.*

We indicate now the number of cells analysed in individual figures and also in Fig. S5C and S8C, respectively.

> *- Fig 5B: What is NP*

Nuclear Pole (NP), also known as the nuclear/acentriolar MTOC (Zeeshan et al 2022; PMID: 35550346).

> *- Fig S1B/D: The legend states that there is an arrow indicating the band, but there is none.*

We have added the arrow.

> *- Fig S2C: Is the scale bar really the same for the zygote and the ookinete?*

We have checked this and used the same for both zygote and ookinete.

> *- Fig S3C, S7C: which stages was qRT-PCR done on?*

Gametocytes activated for 8 min.

> *- Fig. S3D, S7D: According to the figure legend, three independent experiments were performed. How many mice were used per experiment? It would be good to depict the individual data points instead of the bar graph. For S7D, 3 data points are depicted (one in WT, two in allan-KO), what do they mean?*

The bite back experiment was performed using 15-20 mosquitoes infected with WT-GFP and gene knockout lines to feed on one naïve mouse each, in three different experiments. We have now included the data points in the bar diagrams.

> *- Fig S3: Panel letters E and G are missing*

We have updated the lettering in current Fig. S5

> *- Fig 3D: Please indicate what those boxes are. I presume that these are the insets show in b, e and j, but it is never mentioned. J is not even larger than i. Also, f is quite cropped, it would be good to see the large-scale image it comes from to see where in the nucleus these kinetochores are placed. Were there unbound kinetochores found in WT?*

We mention the boxes in the figure legends. It is rare to find unbound kinetochores in WT parasite. We provide large scale and zoomed-in images of free kinetochores in Fig. S8.

> *- Fig S4: Insets are not mentioned in the figure legend. Please add scale bar to zoom-ins*

We now describe the insets in the figure legends and have added scale bars to the zoomed-in images.

> *- Fig S5A, B: Please indicate which inset belongs to which sub-panel. Where does Ac stem from?*

We have now included the full image showing the inset (new Fig. S8).

> *- Fig S5C and S8C: Change "DNA" to "Nucleus".*

We have changed “DNA” to “Nucleus”. Now they are Fig. S8K and S11I.

> ***Reviewer #3 (Significance):***
>
> *Yet, the statement that SUN1 is also important for lipid homoeostasis and NE dynamics is currently not backed up by sufficient data. I believe that the manuscript would benefit from removing the less convincing transcriptomic and lipidomic datasets and rather focus on more deeply characterising the cell biology of the knockouts. This way, the results would be interesting not only for parasitologists, but also for more general cell biologists.*

We have moved the lipidomics and transcriptomics data to supplementary information and toned down the emphasis on these data to make the manuscript more focused on the cell biology and analysis of the genetic KO data.

https://doi.org/10.7554/eLife.106537.1.sa0